

Adapting to Brexit: the Response of Corporate Structures to Geopolitical Uncertainty *

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Abstract

Geopolitical uncertainty alters the incentives of firms to organise their corporate structure across borders, creating a distinct margin of adjustment in response to policy risk. We study this margin using the Brexit referendum as a quasi-natural experiment. We combine firm level data on parent-subsidary links for UK and EU firms between 2011 and 2021 with measures of Brexit-related uncertainty and study changes in foreign subsidiary formation at the extensive margin. Following the referendum, there was an increase in the number of subsidiary formation from the UK into the EU, while the number of EU firms that expanded with subsidiaries into the UK dropped. UK firms establishing their first EU subsidiary after the referendum were systematically weaker ex ante than comparable firms that did so before the referendum. Increased Brexit-related uncertainty is associated with increased foreign subsidiary formation from the UK into the EU, driven primarily by small firms, alongside suggestive evidence of decreased domestic subsidiary incorporation by UK firms. We interpret these findings as evidence of a ‘precautionary’ foreign direct investment channel, operating through changes in the corporate structures of firms in response to geopolitical uncertainty.

KEYWORDS: Brexit, foreign subsidiary, geopolitical uncertainty, parent firm

JEL CLASSIFICATION: F21, F23, G32, F15, D22

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1. Introduction

The June 2016 vote of the United Kingdom to leave the European Union marked the onset of a prolonged period of uncertainty about future market access between the UK and the EU. Firms on both sides of the Channel faced the possibility that the fixed and variable costs of serving a key foreign market might rise through new trade barriers, regulatory divergence, or changes in tax and legal regimes. In such an environment, firms' responses may not be limited to adjustments in prices, quantities, or trade flows. Heightened policy uncertainty may instead induce firms to reorganise their corporate structures, incurring fixed organisational costs to preserve market access and reduce exposure to future frictions. Understanding whether, how, and which firms engage in such organisational responses is central to assessing the real effects of geopolitical uncertainty.

From the announcement of the referendum result in June 2016 and even beyond the conclusion of the Trade and Cooperation Agreement (TCA) in December 2020, uncertainty surrounding the UK-EU economic relationship fluctuated substantially as negotiations evolved. Survey evidence shows that Brexit remained a major source of uncertainty for a significant share of UK firms even after formal withdrawal (Bloom, Bunn, Mizen, Smietanka and Thwaites, 2025). This prolonged and well-defined episode therefore provides a natural setting in which to study how firms adjust their international organisational footprint in response to geopolitical risk. Although our empirical setting is Brexit, the mechanism we study, namely organisational restructuring in response to heightened policy uncertainty, is relevant to a broader set of geopolitical shocks, including trade wars, sanctions, and episodes of regulatory fragmentation (Handley and Limão, 2017).

In this paper, we examine this corporate structure margin by studying how non-financial firms in the UK and the EU responded to Brexit-related uncertainty by establishing new foreign subsidiaries. Using the Brexit referendum as a quasi-natural experiment, we study whether firms expanded their corporate presence across borders to mitigate potential increases in trading frictions and regulatory barriers. We focus on the creation of new parent-subsidiary links as a measure of foreign direct investment (FDI) at the extensive margin, drawing from a comprehensive firm-level dataset covering 13,763 British firms and 10,552 EU firms between 2011 and 2021, from the *Orbis Historical* database.

We document three central findings. First, following the referendum, UK firms continued, and in fact increased, the establishment of subsidiaries in EU member states, while EU firms reduced the pace at which they established subsidiaries in the UK. Second, British firms establishing their first EU subsidiary after the referendum were systematically different from both British firms that expanded to the EU and EU firms expanding within the EU before the referendum. Using difference-in-differences analysis on firm balance sheet data, we show that post-Brexit referendum, these UK firms exhibited lower growth in turnover, profits, sales in the years preceding subsidiary formation. We find the opposite result for EU parents incorporating a subsidiary in the UK. Third, fluctuations in Brexit-related uncertainty are positively and statistically significantly associated with the number of new subsidiaries created by UK firms in the EU and, to a lesser extent, by EU firms in the UK. These effects are economically meaningful, persist for several months, and are driven primarily by small firms.

A key feature of our results is the pronounced asymmetry between UK and EU parents that engaged in international subsidiary incorporations. UK firms faced the risk of losing frictionless access to the EU Single

Market, whereas EU firms retained full access to their largest and most integrated market throughout the period. As a result, the expected benefits of establishing or maintaining a corporate presence inside the EU rose disproportionately for UK firms. Notably, the response to Brexit-related uncertainty is concentrated among small firms, which typically face tighter financial and organisational constraints and therefore have fewer resources to anticipate or gradually implement cross-border relocation. This pattern supports an interpretation of subsidiary formation as an active response to heightened uncertainty rather than the delayed execution of pre-existing expansion plans. In contrast, EU firms expanding into the UK behave more in line with the predicted responses of canonical models of multinational activity, showing stronger balance sheet characteristics and not responding as strongly or persistently to Brexit-related uncertainty.

Beyond statistical significance, the magnitude of the foreign-subsidiary response is substantial. Our estimates imply that one percentage point of the share of firms reporting Brexit as a major source of uncertainty is associated with approximately 23 first-time EU subsidiaries by UK parent firms over a six-month horizon, equivalent to 3.1% of the six-month average number of first new subsidiaries established post Brexit referendum. This translates to an approximate median value of €10.7 million in foreign subsidiary assets. For EU firms expanding into the UK, the analogous cumulative six month increase is smaller, around 12 subsidiaries, i.e. 2.4% of the six-month average number of first new subsidiaries, and €2.6 million in approximate median asset value, consistent with their smaller and less persistent response to uncertainty. These magnitudes indicate that Brexit-related uncertainty induced a meaningful reallocation of organisational activity across borders, rather than a purely marginal response. We also present suggestive evidence that domestic organisational expansion within the UK weakened over the same period. In particular, increases in Brexit-related uncertainty are associated with a decline in the formation of new UK-UK parent-subsidiary links, although these estimates are less precisely measured. While economically small relative to the foreign reallocation effects that we document, this pattern is directionally consistent with existing evidence that Brexit-related uncertainty depressed domestic investment in the UK (Bloom et al., 2025).

This paper contributes to several strands of the literature. Our study contributes to the literature examining the determinants of foreign direct investment by multinational enterprises, such as Brainard (1997), Ramondo and Rappoport (2010), Ramondo, Rappoport and Ruhl (2013), Gumpert, Li, Moxnes, Ramondo and Tintelnot (2020), Anagnostopoulos, Atesagaoglu, Faraglia and Giannitsarou (2022), the growing literature on the impacts of Brexit on cross-border trade and investment, e.g. Crowley, Exton and Han (2018), Mayer, Vicard and Zignago (2019), Breinlich, Leromain, Novy and Sampson (2020), McGrattan and Waddle (2020), Fernandes and Winters (2021), Freeman, Garofalo, Longoni, Manova, Mari, Prayer and Sampson (2025), the literature on the effects of policy uncertainty on firms' decisions, e.g. Kumar, Gorodnichenko and Coibion (2023), Hassan, Hollander, Van Lent and Tahoun (2024), Bloom et al. (2025), and the broader literature on the role of fixed costs on firms' entry into export markets initiated by Melitz (2003).

In particular, this paper contributes to the growing literature on the effects of Brexit and Brexit uncertainty on trade, investment, labour markets and output. Several papers have studied the potential effects of Brexit prior the referendum or shortly after (in reports by HMTreasury (2016) and IMF (2016), as well as Van Reenen (2016), Dhingra, Huang, Ottaviano, Pessoa, Sampson and Reenen (2017), Sampson (2017). More recently, other papers have estimated the effect of Brexit especially on trade and productivity. Among others (Born, Müller, Schularick and Sedláček, 2019; Dhingra and Sampson, 2022; Haskel and Martin, 2023;

Broadbent, Pace, Drechsel, Harrison and Tenreyro, 2024), there are studies focusing on the price and volume of Portuguese exports (Fernandes and Winters, 2021), the entry of British exporters into the EU (Crowley et al., 2018), and the value of UK-EU bilateral trade (Graziano, Handley and Limão, 2020) in the months after the referendum vote find that increases in uncertainty regarding future trade policy led to substantial declines in export activity. Using synthetic control methods, Breinlich et al. (2020) estimate that the Brexit referendum led to a 17% increase in the number of outward investment transactions by UK firms in the EU between 2016 and 2019, Crowley and Hoste (2025) quantify how non-trade-policy barriers between the EU-UK changed following the introduction of the Trade and Cooperation Agreement (TCA), focusing on the bottled water industry. Javorcik, Kett, Stapleton and O’Kane (2025) study the effect of Brexit referendum and related uncertainty on UK job postings, Vanino, Douch and Cakmak (2025) focus on the effects of geopolitical uncertainty on cross-border R&D collaborations and patenting activity after the Brexit referendum, and Breinlich and Magli (2025) study the effect of trade restrictions on services trade using Brexit as exogenous shock. Geiger and Güntner (2024) construct an instrument for Brexit surprises from daily economic policy uncertainty (EPU) and asset-price changes and show that accommodative response of UK monetary policy prevented a stronger negative output effect. We contribute to this literature by providing extensive empirical evidence that higher uncertainty is associated with a reallocation of corporate structures across borders.

This paper is also related to the effects of policy uncertainty on firm decisions. Bloom et al. (2025) study Brexit uncertainty through the creation of a new firm-level survey, the Decision-Maker Panel (DMP). They find that Brexit provided both a major and persistent uncertainty shock, that significantly depressed UK GDP, employment and productivity over the horizon between 2016 and 2025. Hassan et al. (2024) propose a text-based method applied to estimate the impact of Brexit-related uncertainty and find that international firms most exposed to Brexit uncertainty significantly lost market value and also reduced hiring and investments. Kumar et al. (2023) use a new survey of firms in New Zealand and show how exogenous increase in the macroeconomic uncertainty perceived by firms leads to a reduction in their prices, employment, investment, sales and probability that they invest in new technologies or open new facilities. In a different context Bahaj, Piton and Savagar (2024) find that firm entry increased during the COVID-19 pandemic across several countries, and this contrasts with typical recessions where firm entry declines. We contribute to this literature by providing evidence that firms have replaced the creation of new domestic subsidiaries with the establishment of new subsidiaries abroad.

Finally, and from a broader perspective, this paper is related to firms’ growth, exports and foreign direct investment. The seminal work of Melitz (2003) showed how the existence of fixed costs related to exporting lead to only the most productive firms in an economy engaging in export activity. Further research has focused on understanding factors that distort the decision to export (see, e.g. Caggese and Cuñat (2013) and Djankov, Freund and Pham (2010)). The important contribution of Helpman, Melitz and Yeaple (2004) suggests that firms invest abroad when the gains from avoiding trade costs exceed the (fixed) costs of maintaining capacity in multiple markets; further, of those firms that serve foreign markets, only the most productive engage in FDI. Recent work by Fons-Rosen, Kalemli-Ozcan, Sørensen, Villegas-Sanchez and Volosovych (2021) shows empirically how FDI enhances the productivity of acquired firms. Relative to this literature, we contribute by showing that under policy uncertainty this selection mechanism may break down, with

weaker firms responding more strongly due to fears of losing access to export markets.

The remainder of the paper is organised as follows. Section 2 describes the data and variable construction. Section 2.3 presents stylised facts on subsidiary formation before and after the Brexit referendum. Section 3 analyses the characteristics of firms establishing a first foreign subsidiary using difference-in-differences methods. Section 4 studies the dynamic relationship between subsidiary formation and Brexit-related uncertainty. Section 5 presents robustness checks and additional results, and Section 6 concludes.

2. Data Description and Some Stylized Facts

2.1 Orbis data. The core data source is the *Orbis Historical* database, primarily known for its extensive collection of company financial data for millions of firms worldwide, spanning several decades.¹ It is the main source of data that we use to answer our research questions of how British and European firms' corporate structures responded to Brexit as it offers comprehensive data on the ownership structure of entities, and links this with detailed firm characteristics and balance sheet data. Importantly, it provides a representative sample of the European and British economies by including small, unlisted firms (Kalemli-Ozcan, Sorensen, Villegas-Sanchez, Volosovych and Yesiltas, 2024). The good coverage of firm size will allow us to study also any heterogeneous behaviour of firms before and after the referendum, underlining the importance of a holistic analysis of the economy instead of focusing on listed firms only.

To construct our datasets, we use *Orbis Historical* data covering several parent-subsidary configurations. Specifically, we identify British firms with subsidiaries in EU member states, EU firms with subsidiaries in the UK, and EU firms with subsidiaries in EU countries other than the ones that they are incorporated in. We additionally extract data on British firms with subsidiaries in two non-EU countries, Norway and Switzerland, to conduct placebo analyses. Finally, we collect information on British firms with domestic subsidiaries, in order to examine the effect of Brexit-related uncertainty on home-country incorporations. The data extracted are from calendar years 1998 to and including 2021.² Our dataset contains information on the ownership structure of entities, detailing shareholdings, subsidiaries, direct and indirect ownership shares, ultimate owners, and corporate groups.

For our study, we will use the following terms throughout the paper, defined as follows: A *parent* firm is an Orbis shareholder with direct ownership exceeding 0% in a *subsidiary* firm.³ For both parent and subsidiary firms, we have a wide range of firm characteristics including sector, geographic location, date of incorporation, size (Orbis variable), and financial metrics, including turnover, profits, assets, and cash flow.⁴

The main unit of analysis is a *parent-subsidiary link*, which we define as the relationship of a parent firm being the immediate (direct) shareholder of a subsidiary, in a given year of Orbis data. Thus, we keep only

¹We use the Historical Orbis Database for Europe, which comes from Bureau van Dijk (BvD), a Moody's Analytics company.

²Ownership data in *Orbis Historical* starts in 2007, however it is not well populated until 2008.

³Parent firms do not always coincide with *global ultimate owners*, who are defined as firms with indirect ownership of more than 50% of the subsidiary.

⁴Appendix B describes the cleaning procedure for the balance sheet data of firms, following Kalemli-Ozcan, Sorensen, Villegas-Sanchez, Volosovych and Yesiltas (2024).

observations where parent and subsidiary names, subsidiary date of incorporation and direct ownership share are not missing, and we also drop observations reporting a subsidiary's indirect owners, including global ultimate owners. Moreover, to ensure data consistency, we drop parent-subsidiary links where the information date of when the link is formed is missing, or where the year of link predates the year of incorporation of the subsidiary.

We are interested in the establishment of subsidiaries by parent firms, which can be proxied in the data by carefully defining the *formation of a new parent-subsidiary link*. Orbis reports the information date at which the link was reported to them, however, this does not necessarily coincide with the exact date the link was actually formed due to reporting conventions. Instead, the calendar year in which a link first appears in the dataset is a more reliable indicator. We note that the year of a link can be different from the date of incorporation of the subsidiary, for instance, when a parent company just acquires a stake in an existing company.

Since our main research question asks whether and how parent firms establish new, i.e. not pre-existing, subsidiaries in response to Brexit uncertainty, we define a link as *new* if it satisfies the following three criteria: First, the link is labelled as new only for the year that it first appears in the dataset. As an example, consider a parent firm P and subsidiary firm S , and suppose that parent-subsidiary links between P and S exist for years 2013, 2014, 2015 and 2016. Then we count the link in 2013 as a *new* link, since the link was formed in 2013, and interpret this as firm P establishing a new subsidiary in that year. The links in the following years 2014, 2015, and 2016 are recorded in the data as pre-existing, simply reflecting the fact that the parent firm P has established a subsidiary which it keeps owning up and until 2016. Second, we require that the new link is formed within one year of the subsidiary's date of incorporation (DOI) and henceforth use this DOI as our main time variable. This ensures that we truly capture the creation of new subsidiaries rather than parents acquiring stakes in an existing firm. Third, to cleanly identify the formation of a new link, we also condition on the direct ownership share of the parent firm in the subsidiary to exceed 20%.⁵ We refer to a parent firm as a *new parent* and to a subsidiary firm as a *new subsidiary* when the two have formed a new link, as defined above.

We are primarily interested in understanding whether firms with no foreign subsidiaries create new links for the first time after the Brexit referendum. We call such new links between a parent and subsidiary *first new link*, and we formally define it as a new link for which the new parent does not have any other link to subsidiaries (new or pre-existing) in the year prior to the link formation. We refer to parent and subsidiary firms of such first new links as *first-time new parents* and *first new subsidiaries* respectively. By definition, the set of first new links is a subset of the set of new links in our dataset.

We apply two final filters to our baseline datasets. First, we exclude from the samples of parent firms companies that are in the finance and insurance sector. The baseline variable we use for this exclusion is NACE letter code K (financial and insurance activities)⁶. Whenever the NACE letter is not available, we use as much as possible the Orbis variables for trade description and the names of the companies to refine the sector classifications and identify more firms that are in the finance sector. Second, we exclude parent

⁵We can vary the percent direct ownership and the year of link-DOI relation for sensitivity.

⁶According to NACE classification revision 2.

firms for which we have no information about their business activities (either because of lack of sectoral classification, or because we cannot deduce it from text and description variables) and are classified as *holdings* companies.

Applying these data filters and definitions, and restricting the sample to the period from January 2011 to December 2021, covering five years before and after the Brexit referendum in late June 2016, we construct three primary datasets of newly formed parent-subsidary links over time, each with rich firm-level characteristics for both parents and subsidiaries. The first dataset, UK-EU, contains the firm data for all new links of non-financial parent firms headquartered in the UK that establish subsidiaries in EU countries. The second dataset, EU-UK, includes all new links of non-financial parent firms from EU countries that establish subsidiaries in the UK. The third dataset, EU-EU, consists of new links between non-financial parent firms from any EU country that establish subsidiaries in EU countries other than their home country, and thus are classified as foreign subsidiaries. We also construct a placebo dataset, UK-NOCH, which includes new links between UK non-financial parent firms and subsidiaries located in Norway and Switzerland. Finally, we build an auxiliary and substantially larger dataset, UK-UK, containing all new links between UK non-financial parent firms and domestic UK subsidiaries, again with rich firm-level characteristics for both parents and subsidiaries.

2.2 Uncertainty indices and other macroeconomic variables. We gauge uncertainty through the Decision Maker Panel (DMP), a firm-level survey led by a team of researchers at the Bank of England, University of Nottingham and King's College London (Bunn, Bloom, Crundwell, Khan, Menzies, Mizen, Sculthorpe, Shah, Twaites and Yotzov, 2024). The survey results are used to create two uncertainty indexes used in our analysis: (i) Brexit Uncertainty Index (BUI) or (ii) Covid Uncertainty Index (CUI). Each index measures the percentage of firms in each month that identify Brexit/Covid-19 as one of the top three sources of uncertainty in the economy. Spanning 8,000 active UK firms with over 10 employees, the survey employs a rotating, three-panel structure to randomly select firms, ensuring each panel receives one third of the questions monthly and all firms cycle through all questions within each quarter. The BUI data series begins in May 2015, with data prior to September 2016 imputed using betting odds data. The CUI data series starts in March 2020.

In our econometric exercises, we control for several macroeconomic variables at country level that come from several sources. (i) The *Office of National Statistics UK* and *Eurostat* provide the trade data we use as controls in our analysis, specifically variables for exports from UK into EU as whole, as well as from the EU to UK. (ii) the *Bank of International Settlements* provides the real effective exchange rate, (iii) *OECD* provides the real GDP for UK and each European country, (iv) *Fred Data* provides the 3-month interbank interest rate for UK.

2.3 Data summary and stylized facts. Table 1 reports the summary figures for the five datasets we use. The cleaned UK-EU dataset consists of 18,223 new parent-subsidary links, involving 13,763 distinct parent firms and 17,833 new subsidiaries in the EU. Most of the links, 80%, are first time links, where the parent did not have any other subsidiary in the year before the link appeared. More than half (54%) of all new links were formed after the Brexit referendum. Investigating these links over time, we find a large 17.8%

TABLE 1: Summary of Orbis data

	New par-subs links	New parents	New subs	First new par-subs links	First-time new parents	First-time new subs
UK → EU	18,223	13,763	17,833	14,568	13,709	14,303
EU → UK	13,269	10,552	12,899	10,857	10,517	10,557
EU → EU	77,492	58,214	74,297	62,477	58,022	60,081
UK → NOCH	907	850	893	873	850	861
UK → UK	1,218,191	1,025,715	1,181,460	1,040,743	1,024,451	1,027,680

Notes: Period 2011m1-2021m12. Excludes parent firms from sectors ‘financial’ and ‘holdings’.

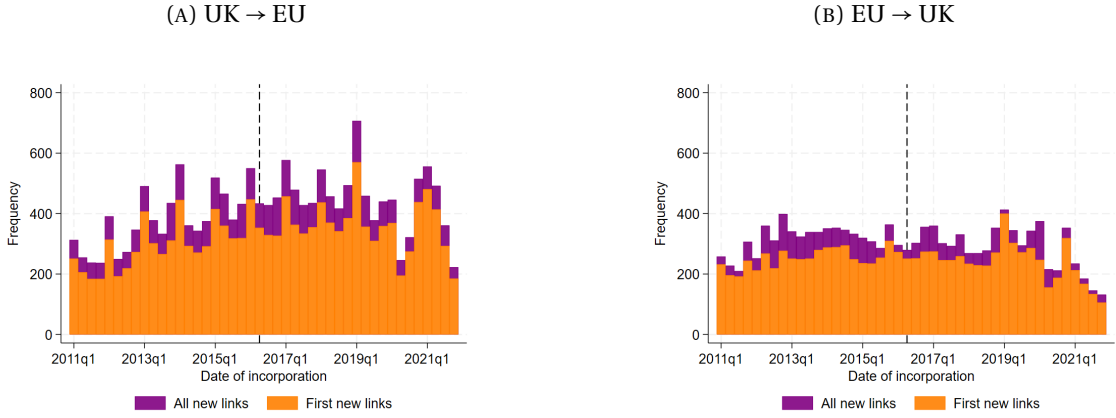
increase in the number of new links formed from before to after the Brexit referendum. This increase is even larger, 20%, when considering only first-time parent-subsidiary links where parents have no prior link in $t - 1$.⁷ Figure 1 panel (A) shows the evolution over time of the number of new parent- subsidiary links at quarterly frequency where the purple bars denote new links and the orange bars first new links. The dotted vertical line identifies the quarter of the Brexit referendum, Q2 2016. Two facts are evident from the figure: first new links are the majority of links that have been created over the sample period accounting over all and across time. Additionally, an upward trend can be identified in the link formation. The referendum does not seem to have broken the upward trend. The only temporary break in the trend can be attributed in the second and third quarter of 2020 during the first period of the COVID pandemic.

Similarly, 10,552 EU non-financial parent firms established 12,899 subsidiaries in the UK between Jan 2011 and Dec 2021, creating 13,269 new links. About 48% of these links are formed post-referendum, a smaller proportion than for UK-EU. Investigating the time series of links, we find the opposite pattern than for UK-EU: The average number of new links formed decreases by 7.8% from before to after the Brexit referendum. Interestingly, this number is slightly smaller for first-time links, decreasing only by 4.4%. This is also evident from Figure 1 panel (B), where our data suggest that the number of links followed a flat or slightly negative trend from 2016 to 2019.

We now turn to the three datasets to be used as placebo or control groups of counterfactual exercises in our empirical analysis. The first is the dataset for links between European parents and European subsidiaries, which is larger, as it encompasses multiple countries on both sides of the relation, with 77,492 new parent-subsidiary links. As such, we find 58,214 distinct non-financial European parents, with 74,297 new subsidiaries are not incorporated in their own country. 53% of these links were formed after the Brexit referendum, an increase of 14.2% from the number of links formed before the referendum. Interestingly, if we zoom in only on first links, the growth rate is lower, 9.4%, and half of the increase in first links created by

⁷ Pre-Brexit refers to 2011q1 to 2016q2, and post-Brexit encompasses 2016Q3 to 2021Q4. We choose the same length in quarters to ease comparison. The time dimension is the subsidiary DOI.

FIGURE 1: New links and first new links



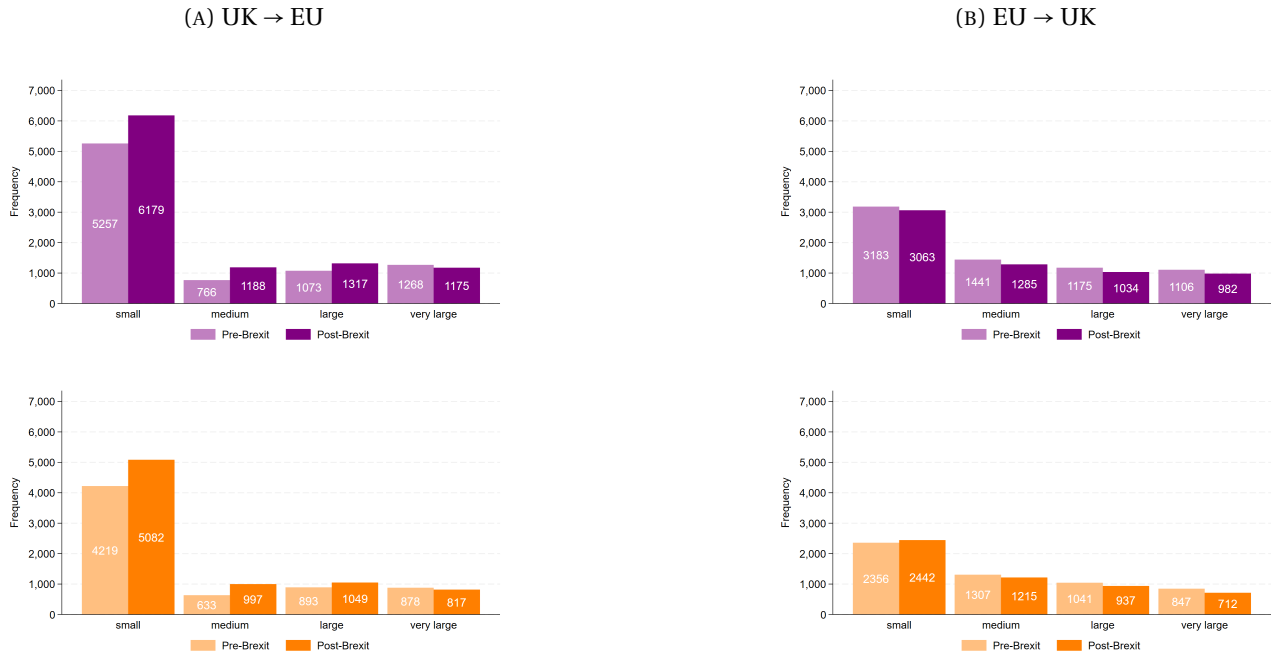
Notes: New parent-subsidary links created by a UK parent in an EU country (panel (A)) and in UK by EU parents (panel (B)). Purple bars denote all new links and orange bars all first new links. The dotted vertical line marks 2016q2, identifying the quarter of the Brexit referendum. Time period 2011q1-2021q4 and frequency is quarterly.

UK parents in the EU. The time series for new incorporations of European subsidiaries by European parents is relatively stable before and after Brexit, showing a positive trend over our sample period (see Figure A.1 panel (A), in the Appendix).⁸ The second dataset includes links between the UK and Norway and Switzerland. There are only 907 links from non-financial British parents to Norwegian and Swiss subsidiaries in our full sample. The number of links decreases substantially from before to after Brexit, -48.3%. This trend holds both for Norway and Switzerland, such that including them in the same sample does not mask any idiosyncratic trends (Figure A.1 panel (B) in the Appendix). The third dataset includes all domestic parent-subsidary links, formed between UK parent firms and UK subsidiaries. It is the largest of all the datasets considered and contains 1,218,191 new subsidiary links, involving 1,025,715 new parents and 1,181,460 new subsidiaries. Of the new links, 1,040,743 are first new links (about 85% of all new links), with 1,024,451 first time new parents, and 1,027,680 first new subsidiaries. We find a decline in link formation following the referendum for this sample (Figure A.1 panel (C) in the Appendix). On average, the number of new links and first new links decreased by 14.1% and 8.5% respectively relative to the pre-referendum period.

We also explore the number of new links by splitting the sample by the size of the parent firm. In Figure 2, the left column shows the UK-EU links by parent size and the right column EU-UK links. The top row shows all new links and the bottom row first new links. First we note that small parents have mostly created new links for the UK-EU, 63%, however this proportion is considerably smaller for the EU-UK dataset, 47%. The UK-EU links of parents of any size have increased after the referendum, apart from the ones created by very large parents. In contrast, we see a decrease in links for parents of any size in the EU-UK sample. If we zoom into the first new links, the picture is similar in both samples. Small UK parents create predominately EU links and medium firms are the ones that show the larger increase after Brexit. Interestingly, the first new UK links of small EU parents are the only ones that slightly increase after the referendum.

⁸For some of the analysis, we use a subset of the EU-EU sample which excludes European firms that also have subsidiaries in the UK, i.e. they appear in our EU-UK sample. There are 1,077 such European firms, i.e. less than 2% of the 58,214 distinct non-financial European parent firms.

FIGURE 2: New links and first new links by parent size



Notes: New parent-subsidiary links created by a UK parent in an EU country (column (A)) and in UK by EU parents (column (B)). Top row shows all new links and the bottom row first new links. Parent size follows Orbis definition. Time period 2011Q1-2021Q4, pre referendum period 2011Q1-2016Q2 and post referendum 2016Q3-2021Q4.

3. Which Types of Firms Establish New Subsidiaries Post-Brexit Referendum?

Our findings so far suggest an increase in subsidiaries established by UK firms in the post referendum period compared to the pre referendum one; in particular, small parents have increased significantly their foreign incorporations, although all size groups have shown an increase. In contrast, we do not see the same pattern in EU-UK incorporations but the opposite both at the aggregate level and when we focus on size groups. In this section, we analyse various characteristics of parents around the time that they decide to establish a subsidiary.

For the rest of the paper we refine our sample in two dimensions. Firstly, we concentrate our analysis to first new links. As discussed in the previous section, first new links are around 80% of all new links in all our samples. This sample restriction will help us study the decision of firms to become multinationals. Second, we exclude from the EU-EU sample any EU firms that, apart from subsidiaries in other EU countries, also have foreign subsidiaries in the UK during our sample period (i.e. they also belong in the EU-UK sample). This restriction allows us to construct a control group that is as much as possible insulated from the uncertainty and structural changes associated with Brexit and more generally the UK.

We start with the premise that typically companies that establish subsidiaries tend to be relatively productive and experience growth (Melitz, 2003; Helpman et al., 2004), and ask whether the uncertainty surrounding future trade policy may prompt a different type of firms to establish a first subsidiary. To investigate this, we focus on the balance sheet characteristics of parent firms in our samples *one* and *two* years prior to establishing their first foreign subsidiary in UK or EU, as indicators of their financial and corporate

‘health’ before their decision to do so.

We conduct a difference-in-difference analysis using repeated cross-sectional yearly data from 2011 to 2021 to assess and compare the health of such UK and EU parent firms before and after the Brexit referendum. To this end, we estimate the following two baseline specifications:

$$Y_{-2,p} = \alpha_{-2} + \beta_{-2}(POST_p \times UKEU_p) + \sum_{j=1}^3 \gamma_{-2}^j X_{-3,p}^j + \varepsilon_p, \quad (1)$$

$$Y_{-1,p} = \alpha_{-1} + \beta_{-1}(POST_p \times UKEU_p) + \sum_{j=1}^3 \gamma_{-1}^j X_{-2,p}^j + \varepsilon_p, \quad (2)$$

where p indexes parent firms. We denote with $Y_{-1,p}$ and $Y_{-2,p}$ the firm dependent variables of interest, as recorded one and two years respectively before the year of incorporation of the first subsidiary of p , in the given sample. The dependent variables we consider are the growth rates of (i) turnover, (ii) sales and (iii) cash flows, and the changes in (iv) return on equity, (v) profits-to-assets ratio and (vi) leverage.⁹ The variable $POST_p \times UKEU_p$ represents the treatment, and is the product of a dummy variable $POST_p$ equal to 1 if the subsidiary is incorporated after the Brexit referendum date, and a dummy variable $UKEU_p$ equal to 1 if the parent firm is in the UK-EU dataset and 0 if it is in the EU-EU dataset. In other words, the treatment variable is 0 if the British parent company established the subsidiary before the referendum, or the parent company is from an EU country and has established a first subsidiary in another EU country before the referendum.¹⁰ We also include three aggregate controls that are lagged relative to the year of incorporation of the first subsidiary and denoted $X_{-3,p}^j$ and $X_{-2,p}^j$ respectively, in order to capture relevant economic conditions in the country of the parent company that may affect the decision of the company to expand with subsidiaries abroad. The controls we include are the growth rates of real GDP, real exports and real effective exchange rate, all for the country of the parent company.¹¹ Finally, ε_p is the residual term and we cluster standard errors at the parent country level.

Table 2 shows the estimated coefficients of interest β_{-2} and β_{-1} from specifications (1) and (2). Across the parent-firm characteristics we examine, our results point to an interesting pattern: UK firms establishing a first subsidiary after the Brexit referendum, when uncertainty about the future trade relationship was high, tended to exhibit weaker performance than non-treated firms. Specifically, such UK firms experienced about 8 percentage points lower turnover growth, and 4 percentage points lower sales growth in the year

⁹The firm balance sheet data is from *Orbis Historical*, all expressed in Euros. For the analysis, we limit our sample to parent firms that report unconsolidated balance sheet data, which constitutes 89% of the sample of UK firms and well over 80% of the sample of EU firms. Prior to doing the regressions, the data was cleaned and prepared following the principles of [Kalemli-Ozcan et al. \(2024\)](#) of how to handle balance sheet data from *Orbis Historical*.

¹⁰In the Figure A.2 in the Appendix, we provide the distribution of the balance sheet variables of interest for the treated and control groups. We do this to first assess if the distribution of dependent variables follows a normal shape, and second to provide an initial understanding of the sources of differences among the groups. Figure A.3 provides the same information for the dependent variables of the EU-UK analysis. The large spikes at the tails of the distributions are due to the fact that variables are winsorised at the 5 and 95 percentiles, as is common when using firm data.

¹¹We note that the variables X are not firm specific, but macroeconomic aggregates. The notation p is used only to map the variable to the country of the parent firm. The sources for all macro aggregate variables we use are *OECD data*, *Bank of International Settlements*, *Eurostat*, and *Office of National Statistics UK*.

TABLE 2: Balance sheet characteristics of UK parent firms with first subsidiary

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-4.146** (1.606)	-8.158*** (1.658)	-7.051*** (1.844)	-3.841** (1.737)
Observations	8,718	9,629	6,628	7,251
R^2	0.002	0.003	0.002	0.001
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	-0.449 (0.319)	-2.248*** (0.405)	1.887 (2.852)	-3.869 (3.703)
Observations	9,398	10,287	8,609	9,371
R^2	0.004	0.002	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.443** (0.175)	-0.230 (0.182)	0.905*** (0.169)	0.274 (0.215)
Observations	10,462	11,501	9,015	9,744
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

preceding their first subsidiary incorporation. These firms were also more leveraged and had a significant drop in their return on equity. Most of these results remain true when we consider the same characteristics two years before the incorporation of the subsidiary, when we see that firms have a decreased profits-to-assets ratio as well. These findings align with the research by [Kumar et al. \(2023\)](#), who show that firms experiencing increased uncertainty tend to reduce employment and witness a decline in sales.

The baseline sample is defined so that the non-treated group consists of EU parent firms that had already incorporated an EU subsidiary before the Brexit referendum only. This choice of control group addresses a potential concern that, following the referendum outcome, some EU firms may have been affected by the shock and responded by expanding into other EU countries rather than the UK. Nevertheless, when we set aside this concern about potential self-selection and include EU parent firms that established an EU link after the referendum as an additional control group, the results remain robust and become even stronger across all dimensions (see Table A.2 in the Appendix, for this full sample of the control group). This suggests that post-Brexit UK parent firms perform even more weakly when compared with this expanded control group.

Next, we repeat the same analysis considering now EU firms that incorporate a subsidiary in the UK after the referendum, using the similarly defined control group as before, i.e. EU parents that have established a first subsidiary in the UK and EU parents that established a first subsidiary in another EU country before

TABLE 3: Balance sheet characteristics of EU parent firms with first subsidiary

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	11.606*** (2.536)	8.197*** (2.247)	12.002*** (2.146)	6.705*** (1.313)
Observations	9,073	10,098	8,306	9,143
R^2	0.004	0.002	0.006	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	3.925*** (1.350)	-1.160 (1.357)	11.409** (4.928)	11.674 (7.267)
Observations	9,916	10,959	9,070	9,899
R^2	0.005	0.002	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	1.450*** (0.509)	-0.209 (0.616)	-0.225 (0.187)	-0.125 (0.326)
Observations	10,950	12,139	9,209	10,003
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

the referendum, estimating the following two specifications:

$$Y_{-2,p} = \alpha_{-2} + \beta_{-2}(POST_p \times EUUK_p) + \sum_{j=1}^3 \gamma_{-2}^j X_{-3,p}^j + \varepsilon_p, \quad (3)$$

$$Y_{-1,p} = \alpha_{-1} + \beta_{-1}(POST_p \times EUUK_p) + \sum_{j=1}^3 \gamma_{-1}^j X_{-2,p}^j + \varepsilon_p, \quad (4)$$

where the notation is the same as in equations (1)-(2). Here, the treatment is the product of $POST_p$, and a dummy variable $EUUK_p$ equal to 1 if the parent firm p is based in an EU country and its first subsidiary is incorporated in the UK. The results reported in Table 3 present a markedly different picture from the previous sample. Most estimates exhibit the opposite sign relative to the earlier exercise and are statistically significant. EU parent firms that incorporated their first subsidiary in the UK after the referendum appear to be healthier than EU parent firms that either undertook the same expansion before the referendum or incorporated a subsidiary in another EU country prior to the vote. When we further expand the control group to include EU parent firms that established subsidiaries after Brexit, the results remain qualitatively unchanged, although somewhat attenuated (see Table A.3 in the Appendix).

Taken together, these findings suggest a clear asymmetry in post-referendum incorporation behaviour. UK parent firms establishing subsidiaries in the EU may have done so primarily as a defensive response to the anticipated frictions associated with Brexit, rather than as part of an expansion strategy grounded in strong expected performance. This is consistent with their weaker pre-incorporation outcomes relative

to comparable EU firms. By contrast, EU parent firms that chose to establish a subsidiary in the UK after the referendum exhibit stronger observable characteristics than relevant control groups, indicating positive selection. Although the overall rate of EU incorporations in the UK declined modestly after the referendum, those firms that did enter the UK market do not appear to be driven by precautionary motives and instead resemble, or even outperform, firms that made similar location choices prior to the referendum.

4. Dynamic Response of First New Links to Brexit Uncertainty

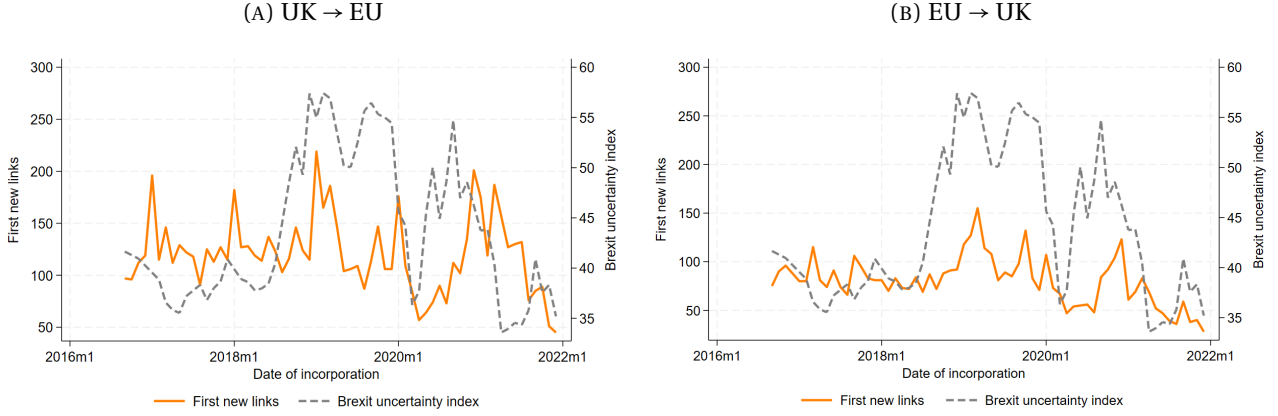
The Brexit referendum introduced an extended period of uncertainty regarding the future trading relationship between the United Kingdom and the European Union. During this time, firms had to weigh the costs and benefits of establishing new subsidiaries across borders while awaiting clarity on post-Brexit trade arrangements. In this section, we examine the dynamic relationship between Brexit-related uncertainty and the monthly number of new subsidiaries established by UK firms in the EU and by EU firms in the UK.

Figure 3 plots the evolution of the Brexit Uncertainty Index (BUI) (right y-axis), i.e. the share of surveyed UK firms reporting Brexit as one of their top three sources of uncertainty, alongside the monthly number of first new links (left y-axis). Panel (A) shows first new links formed by UK firms in EU member states, while Panel (B) shows the first new links by EU firms in the UK. The 2016 Brexit vote triggered a substantial, wide-ranging, and persistent increase in economic uncertainty among UK firms. This uncertainty intensified around major political and negotiation milestones in 2018 and early 2019, and remained elevated throughout 2019 as the UK's departure from the EU was repeatedly delayed. A sustained decline only began after the December 2019 general election and the UK's formal exit from the EU in January 2020. However, Brexit-related uncertainty did not fall consistently below earlier levels until the second quarter of 2021, when the new trade agreement came into force.¹² A visual inspection suggests a positive co-movement between perceived uncertainty and link formation for both UK-EU and EU-UK samples. Periods of increased uncertainty, such as the second half of 2018, coincide with noticeable increases in the number of subsidiary incorporations. The correlation appears weaker in 2020 and 2021, likely reflecting the overlapping effects of the COVID-19 pandemic and the implementation of the final Brexit trade arrangements. Nevertheless, the spike in uncertainty at the end of 2020 is followed by a delayed rise in new EU links by UK firms in early 2021.

To quantify this relationship, we estimate regressions of the monthly number of first new links established by parent firms on lagged uncertainty and macroeconomic conditions. We concentrate on the number of first new links, because it captures more cleanly the extensive margin decision of firms to become multinationals. The time series covers the time period from September 2016 to December 2021, when the

¹²From 2021 onward, uncertainty gradually eased but remained meaningful, affecting roughly 16% of firms, even into 2025, a pattern mirrored by trends in media coverage.

FIGURE 3: First new links and Brexit Uncertainty Index



Notes: First new links over time, for UK-EU sample in panel (A), and EU-UK sample in panel (B), both plotted on the left y-axis, against the Brexit Uncertainty Index, on the right y-axis. Time period 2016m9-2021m12, data sources *Orbis Histotical* and *Decision Maker Panel*, Bank of England.

BUI is available.¹³ For each of the two samples UK-EU and EU-UK, the baseline regression we estimate is

$$Y_t = \alpha + \beta_1 BUI_{t-1} + \beta_2 CUI_{t-1} + \sum_{j=1}^3 \gamma_j X_{t-1} + \delta_1 \mathbf{1}_t(m1) + \delta_2 \mathbf{1}_t(m8) + \sum_{j=2016}^{2021} \zeta_j \mathbf{1}_t(j) + \varepsilon_t, \quad (5)$$

where the dependent variable Y_t is the number of first new links created in month t . BUI_{t-1} is the Brexit Uncertainty Index lagged by one month, and CUI_{t-1} is a similar index, the Covid Uncertainty Index, capturing pandemic-related uncertainty. We include the same three aggregate controls as in the firm level regressions, namely the growth rates of real GDP, real exports and real effective exchange rate, all for the country of the parent firms in the sample. Dummy variables for months January and August, as well as year dummies are included to control for seasonality in subsidiary incorporation and broader temporal trends.

Table 4 reports the main estimates. Columns (1) and (2) show results for UK-EU and EU-UK subsidiary formation, respectively. The coefficient on the Brexit Uncertainty Index is positive and statistically significant in both cases. When we consider UK-EU first new links, a one-percentage-point increase in the share of firms reporting Brexit as a major uncertainty source is associated with approximately three new EU subsidiaries in the following month. The magnitude of the effect is slightly smaller for new incorporations in the UK by EU non multinational parents. While the Brexit Uncertainty Index is constructed using responses from UK firms, it can be interpreted as an upper bound on Brexit-related uncertainty for EU parents. These firms were exposed to uncertainty surrounding the future UK-EU relationship, but continued to benefit from full access to the EU's internal market, implying that their effective uncertainty level may have been lower.

Columns (3) and (4) of Table 4 show the estimated effects for the two placebo samples, EU-EU and UK-NOCH. Column (3) shows no significant relationship between Brexit uncertainty and new EU-EU new incorporations, suggesting that the uncertainty shock was specific to the UK-EU interface. Column (4) confirms that UK first incorporations in Norway and Switzerland (a placebo group) are unrelated to the BUI,

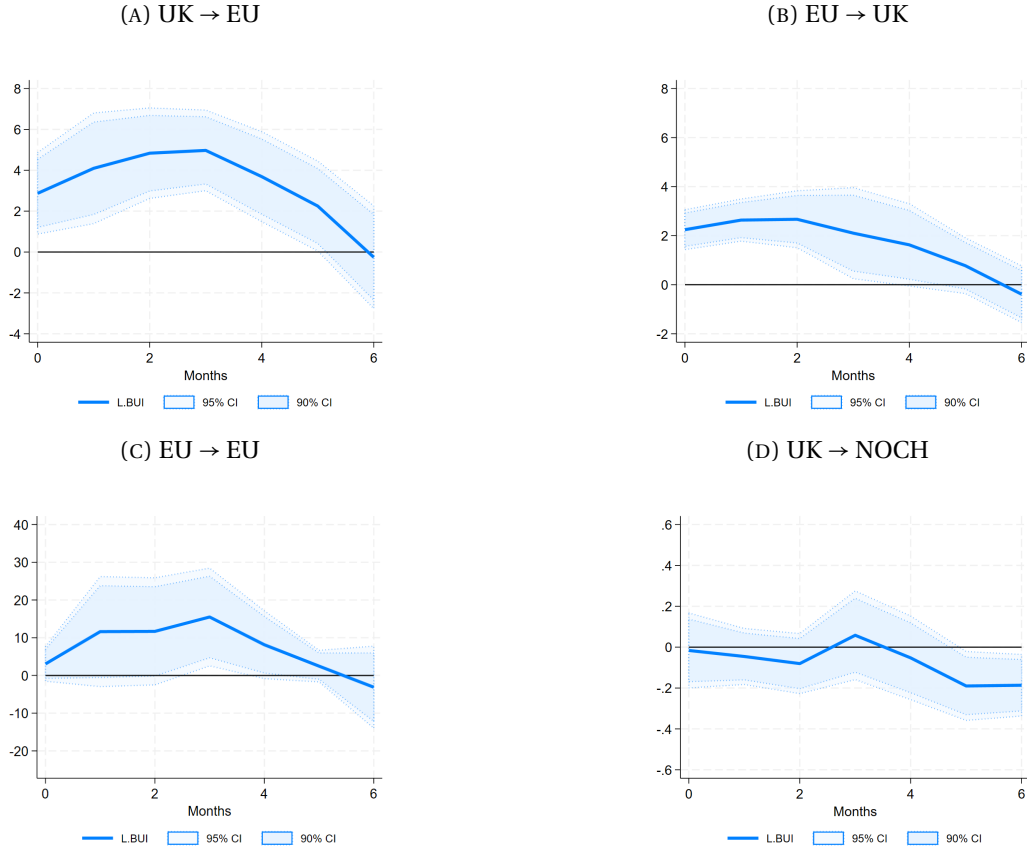
¹³As in Bloom et al. (2025), we use a single measure of uncertainty that will capture simultaneously the first and the second moment effect of Brexit.

TABLE 4: First New Links and Brexit Uncertainty Index

	(1) UK → EU	(2) EU → UK	(3) EU → EU	(4) UK → NOCH
BUI_{t-1}	2.872*** (0.994)	2.245*** (0.406)	3.061 (2.290)	-0.016 (0.091)
CUI_{t-1}	0.371* (0.213)	-0.173 (0.109)	0.537 (0.836)	0.035** (0.014)
gr_{t-1}^{GDP}	0.262 (0.891)	-0.339 (0.868)	-3.359 (5.685)	0.090 (0.063)
gr_{t-1}^{REER}	-0.947 (0.979)	1.948* (0.998)	4.907 (5.718)	-0.026 (0.101)
gr_{t-1}^{EX}	0.922* (0.542)	0.587** (0.289)	4.853** (2.297)	0.000 (0.000)
<i>Constant</i>	-41.656 (50.401)	-14.620 (17.893)	297.966*** (98.175)	5.559 (4.315)
Year dummies	✓	✓	✓	✓
Jan/Aug dummies	✓	✓	✓	✓
Observations	64	64	64	64
R^2	0.600	0.701	0.893	0.268

Notes: Regressions on number of creation of first new subsidiaries (first new links) on lagged Brexit uncertainty BUI_{t-1} , for four samples UK-EU, EU-UK, EU-EU and UK-NOCH, specification (5). All regressions include Jan, Aug month seasonality dummies and year dummies. Monthly frequency, period 2016m9-2021m12. Row 2 shows estimates for CUI_{t-1} . Row 3 shows estimates for lagged real GDP growth rates of the parent firm country. Row 4 shows estimates for real effective exchange rate for the countries of the parent firms. Row 5 shows estimates for lagged growth rates of exports of parent firm country. Time period 2016m9-2021m12, data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Robust standard errors. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

FIGURE 4: Dynamic responses of first new links to uncertainty



Note: Local projection estimates of first new links by parent size on (i) BUI_{t-1} , (ii) CUI_{t-1} , (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummies, (vi) Jan and Aug month dummies (specification 6). Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*.

underscoring that the observed effect is not a general feature of global or European uncertainty.

Covid-related uncertainty, proxied by the CUI, does not appear to influence incorporation behaviour in most of the samples, whereas export growth in the home parent country plays a positive role.

The regressions identify a correlation between one month lagged uncertainty and subsidiary formation. However, the decision by a firm to establish a foreign subsidiary and carry out the implementation of this expansion may take longer. Firms typically require time to gather information, identify a location, assess costs, and complete registration. We therefore estimate the following local projections (Jorda, 2005) to trace the dynamic response of subsidiary creation to uncertainty:

$$Y_t^h = \alpha^h + \beta_1^h BUI_{t-1} + \beta_2^h CUI_{t-1} + \sum_{j=1}^3 \gamma_j^h X_{t-1} + \delta_1^h \mathbf{1}_t(m1) + \delta_2^h \mathbf{1}_t(m8) + \sum_{j=2016}^{2021} \zeta_j^h \mathbf{1}_t(j) + \varepsilon_t, \quad (6)$$

for $h = 0, 1, \dots, 6$ months ahead. Figure 4 presents the estimated effects β_1^h . Panel (A) shows the estimates for the first new links in the UK-EU sample, where we see that the effects of Brexit uncertainty persist for several months, with a peak effect at three months after the reported Brexit uncertainty, before it dissipates

after five to six months. At the peak, a one percentage point increase of BUI_{t-1} is associated with the creation of five (4.97) first new subsidiaries by UK firms. The cumulative effect over six months corresponds to approximately 23 first subsidiaries. To put these numbers in perspective, if 1 out of 100 additional firms surveyed in a given month declared Brexit as one of their top three concerns, this is associated with a six-month cumulative increase of 23 first new subsidiaries. This translates to a 3.1% increase relative to the average six-month number of 736 first new subsidiaries in the period of estimation.¹⁴ Our dataset for the UK-EU allows us to also provide a value calculation of the 6-month cumulative effect of one percentage point of BUI increase, to gauge the value of expansion of foreign investment due to Brexit uncertainty, as follows: the median value of yearly total assets of the subsidiaries in the UK-EU sample is about €468,300, so that the implied cumulative increase in subsidiary value due to one unit of BUI over six months is €10.7 million.

Panel (B) shows the response for EU-UK first new links. The dynamic profile is flatter, peaks at two months and fades after three months, with a cumulative effect roughly half as large, at approximately 12 first new subsidiaries for a one percentage point increase in Brexit uncertainty. A one percentage point increase in the BUI is associated with an approximate 2.4% increase relative to the average six-month number of 492 first new subsidiaries in the period of estimation.¹⁵ In terms of value, the median value of yearly total assets of the subsidiaries of the EU-UK sample is approximately €216,800, which translates into €2.6 million per units of BUI.

For EU-EU incorporations, shown in panel (C) of Figure 4, we do not find strong evidence of a statistically significant dynamic response. However, the estimated effects are positive across horizons and reach weak statistical significance at the three-month horizon. This pattern suggests that uncertainty may have encouraged some EU parent firms to reallocate subsidiary formation within the EU rather than expand into the UK.

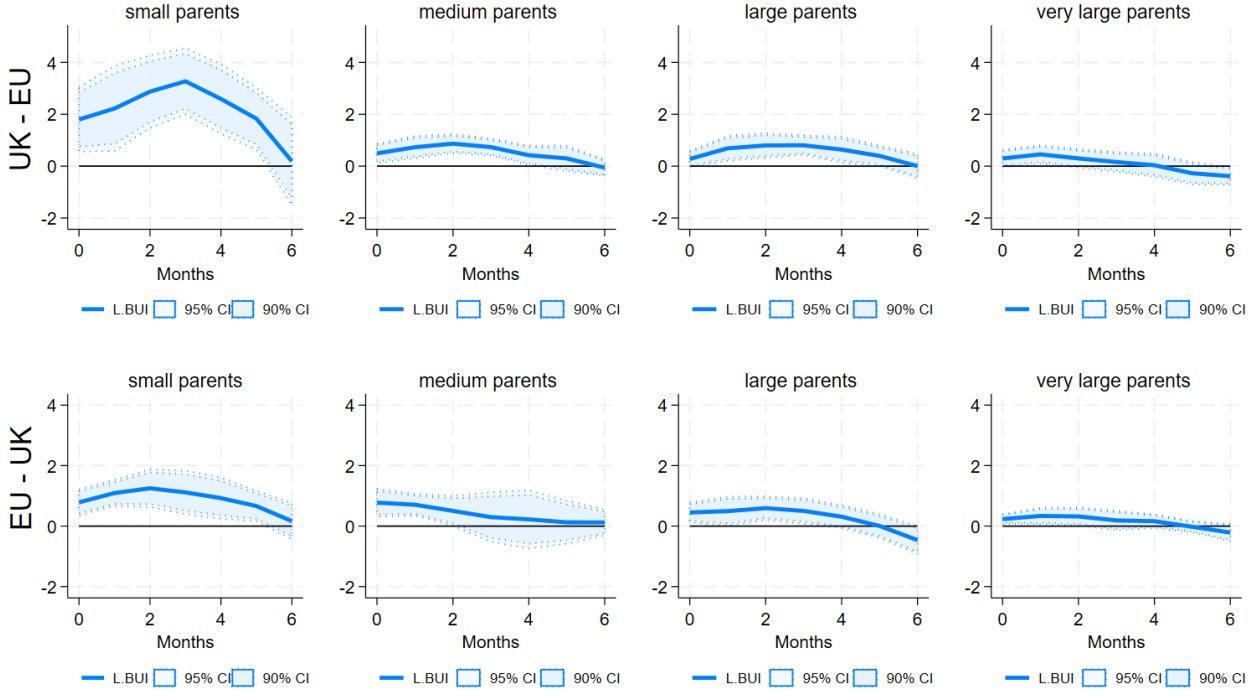
Finally, a placebo analysis using incorporations of new subsidiaries of UK parent firms in Norway or Switzerland, shown in column (4) of Table 4 and 4 panel (D) confirms that these dynamics are specific to Brexit-related uncertainty rather than a general European trend of substitution of markets for UK parents. Incorporations in non-EU European countries were unaffected by fluctuations in the BUI, consistent with the idea that the uncertainty shock was tied to the redefinition of UK-EU relations.

4.1 Results by parent firm size. Next, we explore whether the sensitivity of subsidiary formation to Brexit uncertainty varies by parent firm size. Figure 5 decomposes the impulse responses using the *Orbis* size classification. The top row shows the dynamic responses for the UK-EU sample and the bottom row shows the responses for the EU-UK sample. The results suggest that small parent firms, shown in the first column of the figure, account for the bulk of the response. Among UK parents, the number of first new links for small parent firms has the largest and most persistent reaction, with the peak occurring three months after the uncertainty shock. Medium and large firms also respond, but the cumulative effects are less than half as large, while very large firms display only short-lived and statistically insignificant responses. The results in the bottom row of Figure 5 confirm that small EU parents have also been more responsive to

¹⁴If we consider the 90% confidence intervals, the range of percent increase is 1.3% to 4.8%.

¹⁵Considering again the 90% confidence intervals, this translates to an increase in the range between 0.9% and 3.8%.

FIGURE 5: Dynamic responses of number of first new links to uncertainty, by parent size.



Notes: Top row shows IRFs for UK-EU sample, and bottom row shows IRFs for EU-UK sample. LPs of first new links by parent size on (i) BUI_{t-1} , (ii) CUI_{t-1} , (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummy variables, (vi) Jan and Aug month dummies. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical*, *Office of National Statistics UK*, and *Eurostat*.

Brexit uncertainty than bigger firms, but the response has been flatter and more short-lived. Cumulatively, the response of the number of first new subsidiaries by small parent firms accounts for approximately two thirds (65.8%) of the overall response for the UK-EU sample and approximately half (51.4%) of the overall response for the EU-UK sample. This heterogeneity highlights that smaller firms, which typically face higher fixed costs of cross-border operations, appear to have been more proactive in mitigating Brexit-related risks through the establishment of EU subsidiaries.

4.2 Establishing subsidiaries domestically. Our results so far point to an increase in the creation of new foreign subsidiaries, both from the UK into the EU and from the EU into the UK, in response to increases in Brexit-related uncertainty. This pattern suggests that firms adjusted their international organizational structures as uncertainty rose, potentially as a way to preserve market access, manage regulatory risk, or reallocate activities across borders. We now extend this analysis by complementing and contrasting these findings with evidence on the domestic formation of parent-subsidiary links within the UK.

A growing body of evidence indicates that Brexit uncertainty has been associated with a marked contraction in UK investment overall. Bloom et al. (2025) estimate that aggregate investment declined by between 12% and 16% by 2025 following the referendum, with the negative effect accumulating gradually over time

and uncertainty accounting for a substantial share of the overall decline. Similarly, Keiller (2024) documents a 7% reduction in investment in the UK manufacturing sector in the post-referendum period. Taken together, these findings suggest that, while some firms may have responded to Brexit uncertainty by expanding their foreign footprint, domestic investment activity within the UK weakened.

Against this backdrop, we turn to our final dataset, UK-UK, which records all parent-subsidary links formed between UK parent firms and UK subsidiaries over the period January 2011 to December 2021. Our earlier analysis focused on the response of foreign investment to Brexit uncertainty. By the same token, the evolution over time in the number of (first) new links between UK parents and UK subsidiaries can be interpreted as a proxy for home investment activity. The UK-UK dataset therefore allows us to estimate directly the impact of Brexit-related uncertainty on domestic organizational expansion within the UK.

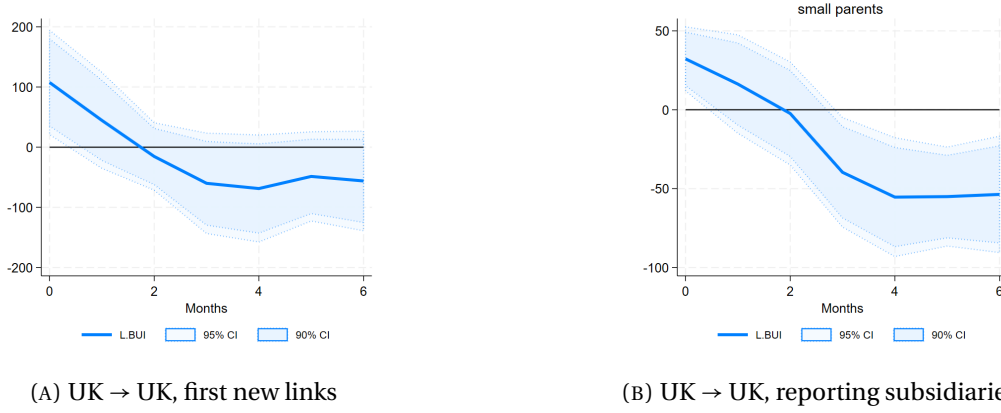
Panel (A) of Figure 6 presents the local projection estimates based on the sample of first new UK domestic links. The cumulative response is negative throughout the horizon considered, and the direction and magnitude of the estimated effect are consistent with the adverse impact of Brexit uncertainty on domestic investment documented in the existing literature. We note that the decline in the number of domestic subsidiaries primarily comes from small parent firms that refrain from expanding within the UK: Panel (B) shows the number of first new domestic links, for small UK parent firms, when we consider the subsample in which each first subsidiary reports some business activity, in the form of total assets or operating revenue, at least once between the one year before and one year after the link with the parent firm.¹⁶

To facilitate comparison with our estimates for foreign subsidiary formation in the UK-EU and EU-UK cases, we also compute that a one percentage point increase in the BUI is associated with a cumulative reduction of 97 new domestic subsidiaries over a six-month horizon. Given a median value of annual total assets of approximately €11,300 for UK subsidiaries in our sample, this translates into a reduction in domestic asset creation of around €1.1 million per percentage point increase in BUI. This contrast between declining domestic link formation and rising cross-border subsidiary creation underscores the reallocation dimension of firms' responses to Brexit-related uncertainty.

Overall, the evidence in this section shows that Brexit-related uncertainty did not simply depress investment activity, but reshaped it where firms chose to expand. Increases in uncertainty were associated with a persistent rise in cross-border subsidiary formation, particularly from the UK into the EU, alongside a weakening of domestic subsidiary creation within the UK. This asymmetric response reflects firms' differential exposure to the risk of losing frictionless access to the EU Single Market: UK firms faced the prospect of heightened trade and regulatory barriers, whereas EU firms retained access to their larger domestic market. The response was economically meaningful, unfolded over several months, and was concentrated among smaller firms, which typically have fewer alternative channels for maintaining market access. Rather than waiting for policy resolution, firms responded pre-emptively by reorganising their corporate structures across borders. We interpret this pattern as evidence of *precautionary FDI*, whereby firms reallocated organisational expansion internationally as a hedge against future trade and regulatory frictions.

¹⁶This is done in order to capture actual, productive activity by subsidiaries, rather than 'paper' or 'shell' entities. See more details on this in Section 5.

FIGURE 6: Dynamic response of domestic first new links to uncertainty



Notes: LPs of first new links by parent sector on (i) BUI_{t-1} , (ii) CUI_{t-1} , (iii) macro controls (growth rate of real GDP of UK, real effective exchange rate of UK, 3-month interbank rate) (v) year dummy variables, (vi) Jan and Aug month dummies. Panel (A) shows estimates for number of first new subsidiaries; Panel (B) shows estimates for number of first new subsidiaries of small parents, restricting to sample of subsidiaries that report balance sheet data. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical*, and *Office of National Statistics UK*, *FRED data*, and *Bank of International Settlements*.

5. Robustness and Further Results

The results are robust to a wide range of sample refinements. We first present the robustness exercises for the difference-in-differences (DiD) estimations and then those for the local projections (LP) estimations, in relation to the foreign subsidiary activity of our two main samples, UK-EU and EU-UK. In both cases, we report results for several carefully considered subsamples of the data, as we explain below. For reference, Table A.1 in the appendix reports the information on the subsamples we consider.

5.1 Balance sheet characteristics of parent firms that establish foreign subsidiaries. We repeat the DiD analysis on three refinements of the samples: (i) one that consists of parent firms that established their first and *only* subsidiary during the sample period 2011-2021, (ii) one that excludes the years 2016 and 2017 and (iii) one that excludes parent firms from an unknown sector. Below we describe each of the subsamples, and if and how the results change relative to the samples of first new subsidiaries.

First, our baseline sample consists of firms that created their first new link within the sample period but some of these firms may have created more than one link during this time. For example, a first subsidiary in 2016 followed by a second one in 2018. For these firms, the decision to expand into a multinational structure with multiple subsidiaries may have been driven by a combination of both Brexit-related considerations and other business and operational motives. In contrast, firms that became multinationals by establishing only one subsidiary during the period are more likely to have been influenced primarily by the uncertainty induced by Brexit. This restricted sample accounts for 79% for the UK-EU sample and 88% for the EU-UK sample of the first new links in the corresponding baseline samples. Table A.4 reports the results for UK parent firms creating their first and only link in the EU. These parents exhibit even weaker performance across several dimensions, such as growth in turnover, sales, as well as changes in leverage, than the parents

in the baseline sample reported in Table 2. This pattern suggests that firms opening only one subsidiary after the referendum were not scaling up an already strong business, but rather engaging in a precautionary response to heightened uncertainty. Consistent with the baseline estimations, the results for EU parent firms that incorporated their first and only subsidiary in the UK, presented in Table A.5, reveal no significant difference than the results presented in Table 3.

Second, the years 2016 and 2017 were characterised by a sharp depreciation of the British pound following the referendum outcome, which may have adversely affected the financial position of parent firms in the sample. Although the baseline specification controls for changes in UK competitiveness using the UK real effective exchange rate, we further assess robustness by restricting the sample. Tables A.6 and A.7 in the Appendix report the corresponding results and confirm the robustness of the findings. The results for both specifications are in line with our baseline estimates.

Third, we exclude from the sample any parent firms whose sector is classified as ‘unknown’ by our procedure, which reduces the sample to approximately 69% of the original set of first new links. Excluding these firms is potentially important because this category may include financial subsidiaries or holding companies that cannot be reliably identified during the classification and data-cleaning process. We note that a larger share of parent firms fails to report sector information in the UK than in the EU, as shown in Table A.1. The corresponding estimates are reported in Tables A.8 and A.9 for the UK and EU, respectively, and are robust to this difference in reporting. The estimated coefficients are very close to those in the baseline specification, confirming that UK firms establishing their first subsidiaries in the EU after the referendum exhibited weaker pre-establishment financial health relative to firms that did so before the referendum.

5.2 Dynamic response to Brexit uncertainty. We next turn to robustness exercises for the local projections. In particular, we assess the robustness of the results using alternative samples from two groups. First, with respect to the characteristics of the parent firms, we consider (i) the sample of *all* new links formed between 2016 and 2021, (ii) the sample with links that are the first and only links of the parent firm, and (iii) the sample that excludes links of parents of ‘unknown’ sector. Second, with respect to features of the subsidiaries, we consider (iv) the sample that excludes links of subsidiaries classified as financial sector, holding company or unknown sector and (v) the sample that excludes links with subsidiaries that do not report basic balance sheet data, e.g. total assets or operating revenue. The local projection estimates for (i)-(iii) are shown in Figure A.4, and the estimates from (iv)-(v) are shown in Figure A.5, both in the Appendix.

The first row of A.4 shows the estimates based on the full sample of new links, which is not limited to first links but also includes any new link formed by a parent firm. As a result, at the time of incorporation, parents may already have links to other subsidiaries, meaning that this sample also includes firms that are already multinational. Consequently, this specification does not capture only the extensive margin of the decision to become a multinational. The qualitative pattern of the baseline results remains unchanged, although the magnitude of the effects is marginally larger. In particular, for the UK-EU case, the maximum correlation between the BUI and new subsidiary links still occurs three months after the increase in the BUI, but it is larger and closer to 6. The cumulative effect after six months amounts to a 3% increase relative to the average six-month number of 912 new subsidiaries in the period of estimation, per one percentage point of BUI. For the EU-UK case, the contemporaneous (month 0) correlation is almost 1.5 times larger, and it

displays a clear declining pattern over the six-month horizon. This indicates that the baseline findings are not driven by the restriction to first-time multinational activity and that heightened uncertainty affects both the extensive margin of becoming a multinational and the subsequent expansion decisions of firms that are already internationally active.

The second row of Figure A.4 reports the local projection estimates for the subsample of parents with a first and only one subsidiary in the period 2011-2021. Our baseline results are also robust to this variation and in line with the corresponding DiD robustness exercise, as confirmed by the responses in the figure.

As an additional robustness check, we again verify that the results are not driven by parent firms whose sector is classified as ‘*unknown*’. The last row of Figure A.4 reports the local projection estimates when restricting the sample to parents with a known sector classification. The results remain qualitatively unchanged for both the UK-EU and EU-UK samples. In particular, we continue to observe a delayed and persistent response in EU subsidiary links formed by UK parents, alongside a flatter response in UK subsidiary links formed by EU parents. Quantitatively, the effects also remain in both directions, although the peak response in the UK-EU sample is slightly attenuated.

Finally, to ensure that our results are not driven by parents not being the ultimate owner of the subsidiaries, we restrict our sample to the links where parents are also listed as the global ultimate owner (GUO) of 25% and of 50% of the subsidiary. This excludes, for instance, a link between a British parent firm with an EU subsidiary whose eventual majority owner is any firm other than the same British parent, and thus controls for indirect ownership. We note that the GUOs of 25% are not a perfect subset of the GUOs of 50%, and that our imposed restriction is the most conservative one. This reduces our UK-EU and EU-UK samples to a third of the baseline, with 6,087 distinct links for UK-EU, and 4,762 distinct links for EU-UK (Table A.1). Figure A.4 shows that our results remain qualitatively unchanged.

We now turn to the subsidiary characteristics exploiting their sector and their balance sheet figures. A first concern is that a large share of newly created subsidiaries may belong to the financial sector. In particular, the EU-UK results could potentially be driven by EU parent firms seeking to maintain access to British financial services. To address this, we repeat the local projection estimation after excluding all links for which the subsidiary operates in the financial sector or the sector is unknown, or classified as a holding company. The resulting samples therefore include only links in which both parent firms and subsidiaries are non-financials and represent 62.1% of the baseline UK-EU sample and 68.8% of the EU-UK baseline sample. The first row of Figure A.5 shows that the baseline results are virtually identical when using this restricted sample, indicating that access to financial markets is not driving the results.

A second concern is whether the subsidiaries established by parent companies are entities with no real business activities, and exist only on paper. Our dataset does not allow us to strictly identify such potential ‘*shell*’ subsidiaries, because both in the UK and EU it was not a legal requirement for small entities to report detailed balance sheet data during the time period of our samples. Therefore, lack of balance sheet data entries can mean either that the company is not reporting it or that the company has no business activity. We can nevertheless restrict the sample to contain first subsidiaries that report some business activity in the period that they are linked to a parent firm, and consider this sample a proxy for excluding such paper-based companies. Specifically, we require that each first subsidiary in our samples reports some business activity in the form of total assets or operating revenue, at least once between the one year before and one year after

the link with the parent firm. With this refinement, the number of links is approximately halved for the UK-EU sample and drops by one third for the EU-UK. Despite that, as shown in the second row of Figure A.5, the local projection estimates continue to exhibit the same qualitative features as in the baseline, albeit at lower levels for both UK-EU and EU-UK samples. This evidence supports the view that the response of British and European firms that established subsidiaries is attributed to fear of losing access to foreign trade markets rather than tax evasion or other illicit activity.

5.3 Response to Brexit uncertainty at sectoral level. Our data further allow us to explore heterogeneity in firms' responses by focusing on the sector of the parent company. Table A.10 shows the breakdown of sectors for the UK-EU and EU-UK samples, excluding financial firms and holding companies (using the same classification procedure as with the other samples). Figures 7a-7b report the local projection estimates of the effects of BUI_t for the UK-EU and EU-UK samples, respectively, disaggregated by the sector of the parent firm that established a first subsidiary link between 2016 and 2021.¹⁷ This sectoral breakdown provides additional insight into which types of firms are primarily driving the aggregate patterns documented in the baseline results.

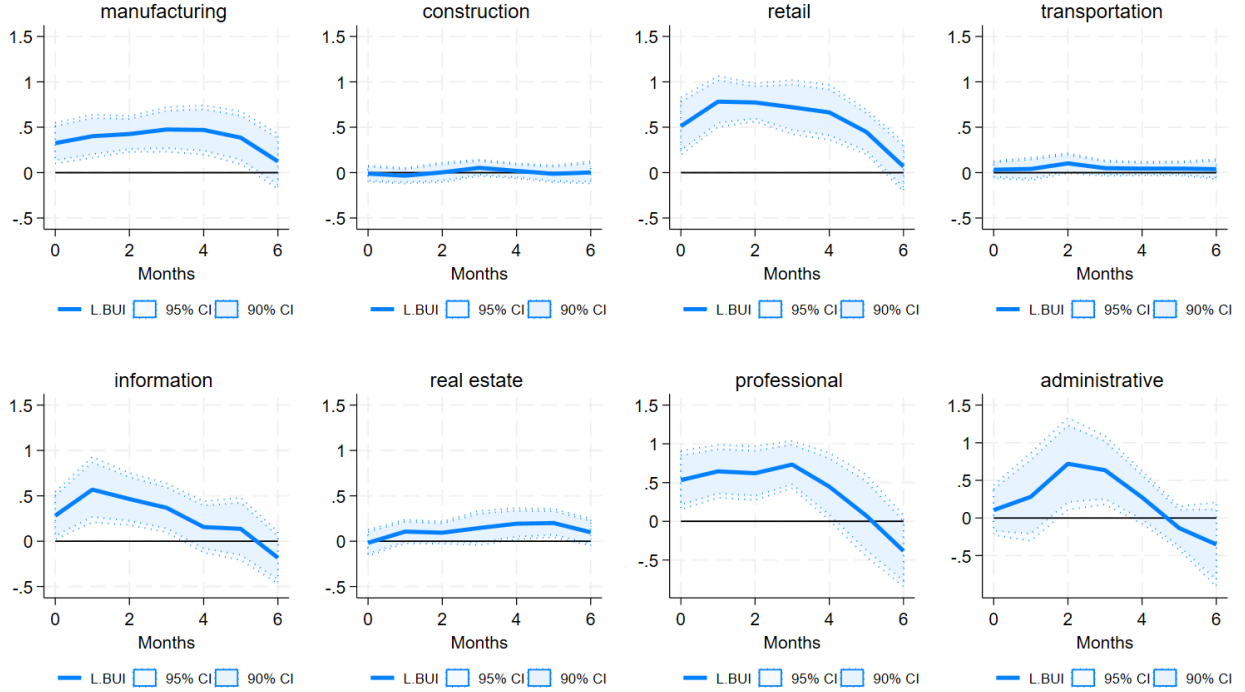
For the UK-EU sample, five sectors account for the bulk of the overall effect: manufacturing, retail, information, professional services, and administrative activities. Among these, the responses of the retail, professional, and information sectors are particularly pronounced, both in terms of magnitude and persistence. These sectors exhibit a clear and sustained increase in subsidiary formation after increased uncertainty, suggesting that firms operating in activities that rely heavily on cross-border market access, client proximity, and regulatory continuity were especially responsive. This pattern is consistent with anecdotal evidence surrounding Brexit-related corporate restructuring and is in line with existing findings for service-sector firms, as documented by Breinlich and Magli (2025).

In contrast, the sectoral responses in the EU-UK sample are generally more muted, mirroring the weaker aggregate effects observed in the baseline analysis. Nevertheless, two sectors stand out: professional services and information. The responses in these sectors suggest that EU-based firms in knowledge-intensive and client-facing activities were more likely to establish a UK presence in response to increased uncertainty. This is particularly indicative of professional service providers, such as legal, consulting, and advisory firms, for whom maintaining direct access to foreign markets and clients is critical. Overall, the sectoral evidence supports the interpretation that these location decisions were shaped by firms' efforts to preserve operational continuity and access to key markets in activities that are highly sensitive to regulatory and institutional changes.

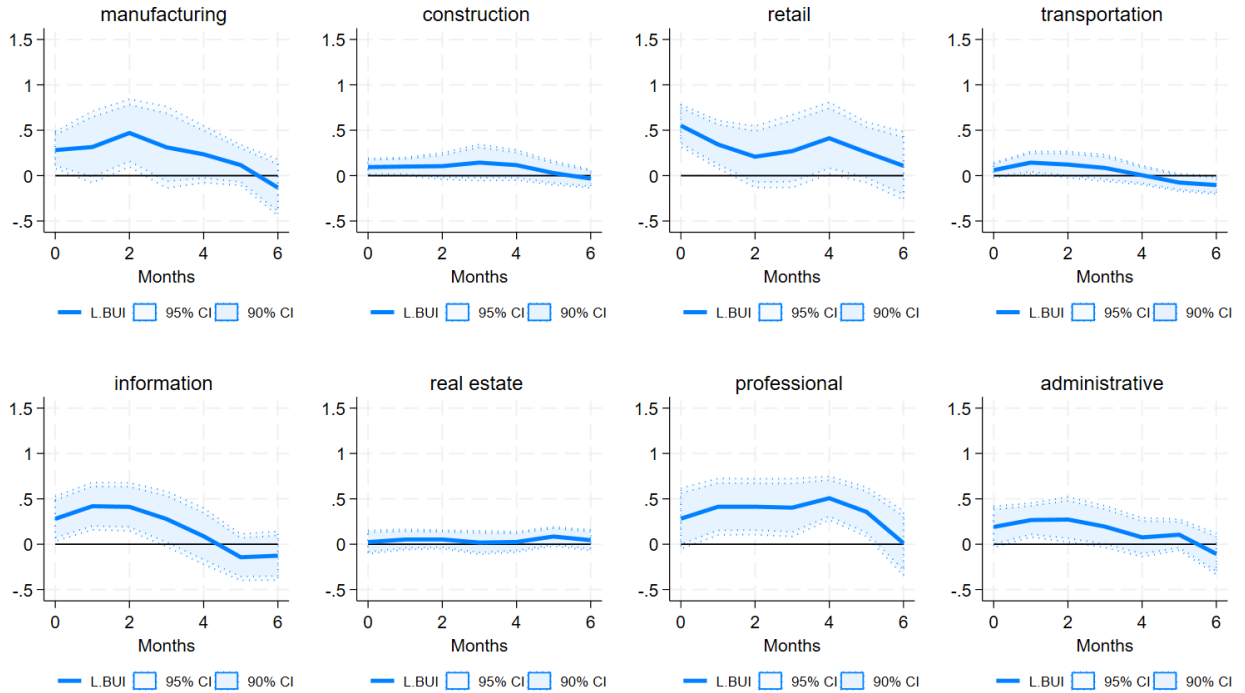
¹⁷We only present the estimated effects for the eight largest sectors in our samples, after excluding financial sector and holding companies. Using NACE classification rev. 2, these are *manufacturing* (C), *construction* (F), *wholesale and retail trade* (G), *transportation and storage* (H), *information and communications* (I), *real estate activities* (L), *professional, scientific and technical activities* (M), and *administrative and support services* (N).

FIGURE 7: Dynamic Responses of first new Links to Uncertainty by Parent Sector

(A) UK → EU



(B) EU → UK



Note: Top panel (A): UK-EU sample and bottom (B): EU-UK sample. LPs of first new links by parent sector on (i) BUI_{t-1} , (ii) CUI_{t-1} , (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummy variables, (vi) Jan and Aug month dummies. Time period 2016m9-2021m12. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources Orbis Historical, Office of National Statistics UK, Bank of International Settlements and Eurostat.

6. Conclusion

This study examines the multifaceted effects of Brexit on the internationalisation strategies of British and European firms, focusing on decisions to establish subsidiaries in the EU and the UK between 2011 and 2021. The June 2016 referendum marked the onset of an unprecedented period of uncertainty regarding future market access, border frictions, regulatory alignment, and tax regimes. In response, firms on both sides of the Channel were compelled to reassess their organisational structures and investment strategies. Against this backdrop, we analyse whether and how the creation of subsidiaries served as a strategic instrument to mitigate anticipated trade frictions following the UK's exit from the EU Single Market.

Our empirical analysis yields three central findings. First, we document an increase in the incorporation rate of UK-owned subsidiaries in the EU alongside a decline in EU-owned subsidiaries in the UK. These diverging patterns point to heterogeneous motives underpinning multinational expansion and suggest that British firms, in particular, intensified their use of EU subsidiaries as a defensive response to the anticipated loss of seamless market access. Second, using a difference-in-differences framework, we show that British parent firms establishing their first subsidiary in the EU after the referendum experienced significantly weaker performance, measured in terms of turnover, profits, assets, and cash flow, among others, relative to comparable UK firms that internationalised prior to the referendum and to EU firms expanding within the EU over the same period. In contrast, we find evidence that EU parent firms establishing their first subsidiary in the UK exhibit systematically better balance sheet outcomes relative to their control group. This asymmetry highlights the disproportionate adjustment costs borne by British firms in the post-referendum environment. Third, our analysis of temporal dynamics reveals a strong and persistent positive relationship between the Brexit Uncertainty Index and the formation of links between British parent firms and EU subsidiaries. While EU-UK subsidiary links are also positively associated with uncertainty, the effect is smaller and less persistent. These findings are robust for several sample refinements and provide compelling evidence that firms may have deployed subsidiaries as a risk-mitigation device in periods of heightened policy uncertainty, which we can label *precautionary FDI*.

Our paper contributes to the literature on foreign direct investment, policy uncertainty, and the firm-level consequences of major geopolitical shocks. By documenting both performance effects and strategic reallocation responses, we shed light on how firms adapt organisationally to abrupt changes in the institutional environment. Beyond the immediate Brexit context, our findings underscore the broader role of uncertainty in shaping multinational investment decisions.

Several avenues for future research emerge from our analysis. First, assessing the long-term viability and performance of post-Brexit EU subsidiaries established by British firms would deepen our understanding of whether precautionary FDI translates into sustainable growth. Second, exploring heterogeneity between industries, regions, and firm characteristics could clarify which firms are most exposed to and most capable of adapting to geopolitical disruptions. Finally, further work that examines the interaction between regulatory divergence, trade agreements, and multinational firm strategies would help situate Brexit within the evolving architecture of global trade and investment. Together, these extensions would contribute to a more comprehensive understanding of how firms navigate uncertainty in an increasingly fragmented global economy.

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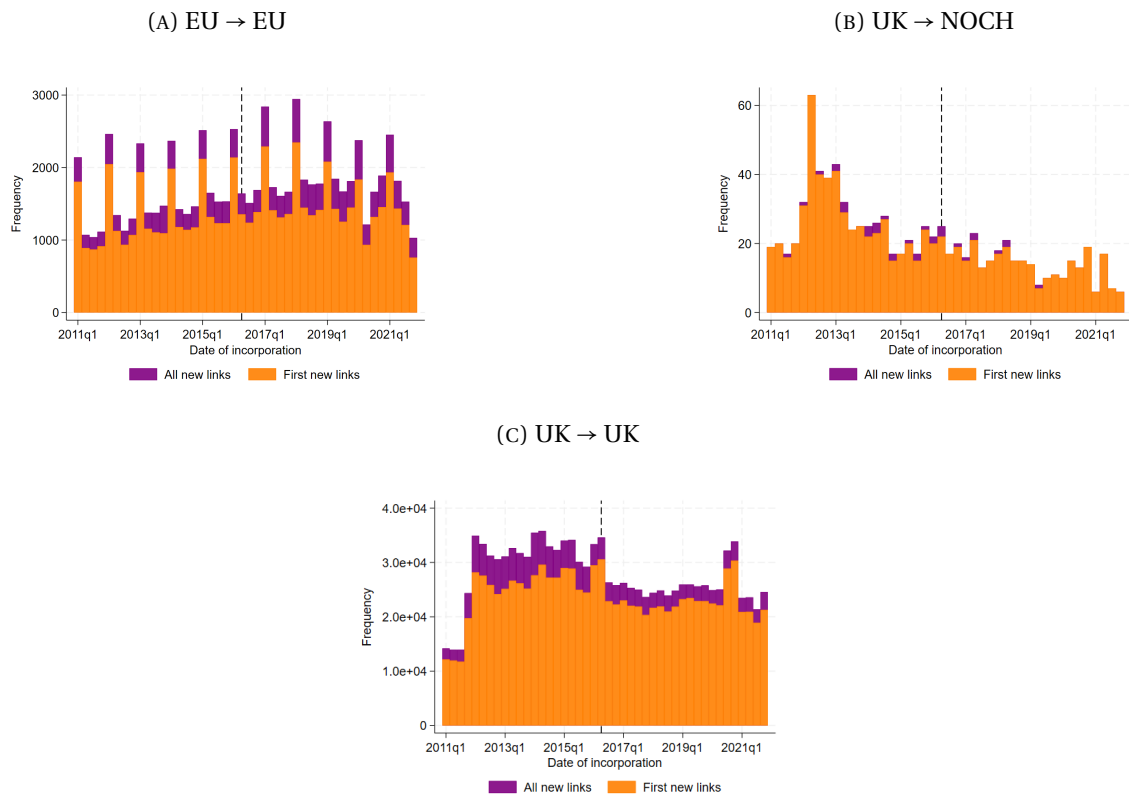
A. Additional Facts and Results

TABLE A.1: Time series of new links frequencies

	UK-EU	EU-UK
All new links	18,223	13,269
First new links	14,568	10,857
First ever new links	13,384	10,314
First and only new links	11,473	9,506
First links, no unknown parents	10,003	9,431
First links, no unknown subsidiaries	13,732	10,030
First links, no financial subsidiaries	12,160	9,963
First links, no financial or unknown subsidiaries	11,324	9,136
First links, subsidiaries with balance sheet records	9,601	8,993
First links, parents are also GUO (25%) and GUO (50%)	6,087	4,762

Notes: Period 2011m1-2021m12. Excludes parent firms from sectors 'financial' and 'holdings'. GUO refers to global ultimate owner. Data source *Orbis Historical*.

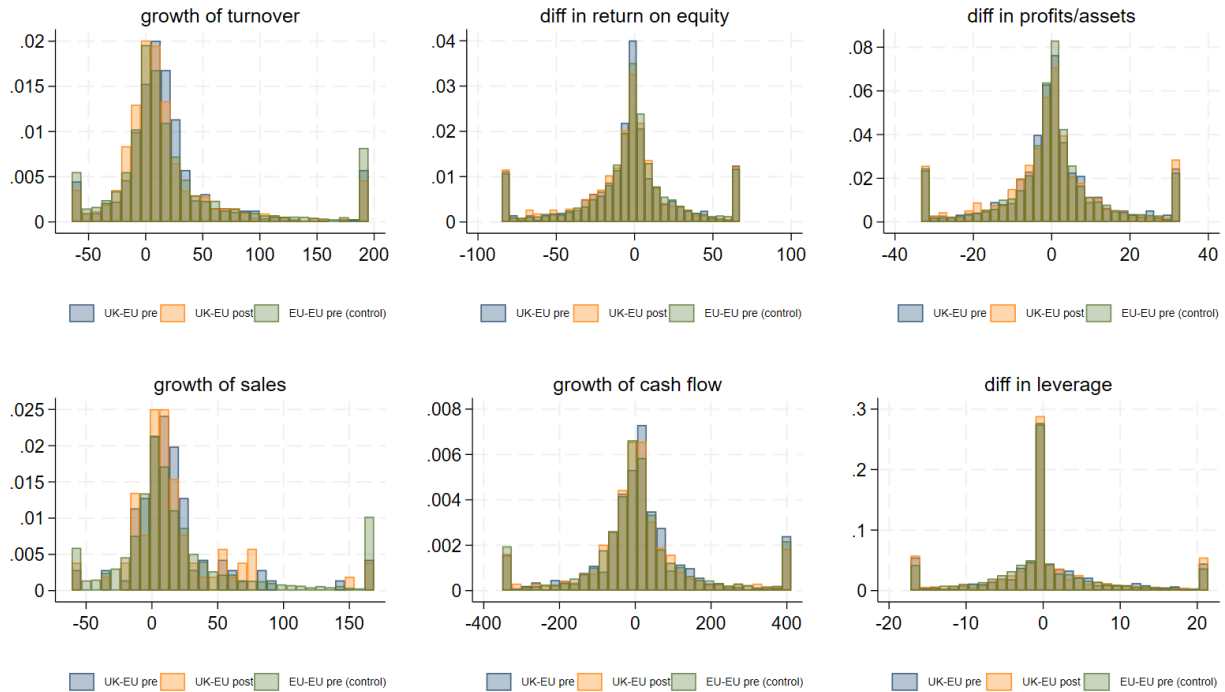
FIGURE A.1: New links and first new links



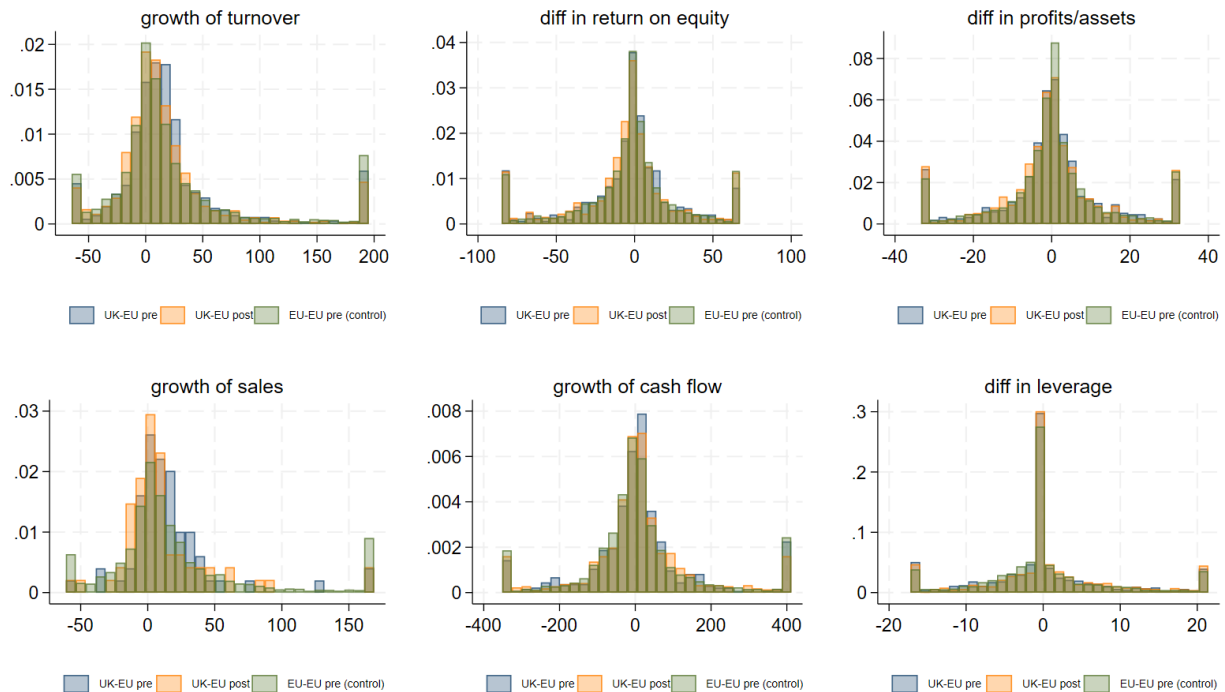
Notes: New parent-subsidary links created by a EU parent in a EU country (panel (A)), by a UK parent in Norway or Switzerland (panel (B)), by a UK parent in UK (panel (C)). Purple bars denote all new links and orange bars all first new links. The dotted vertical line marks 2016q2, identifying the quarter of the Brexit referendum. Data source *Orbis Historical*, time period 2011q1-2021q4.

FIGURE A.2: Balance sheet distributions of parent firms, sample UK-EU

(A) One year before the first subsidiary incorporation



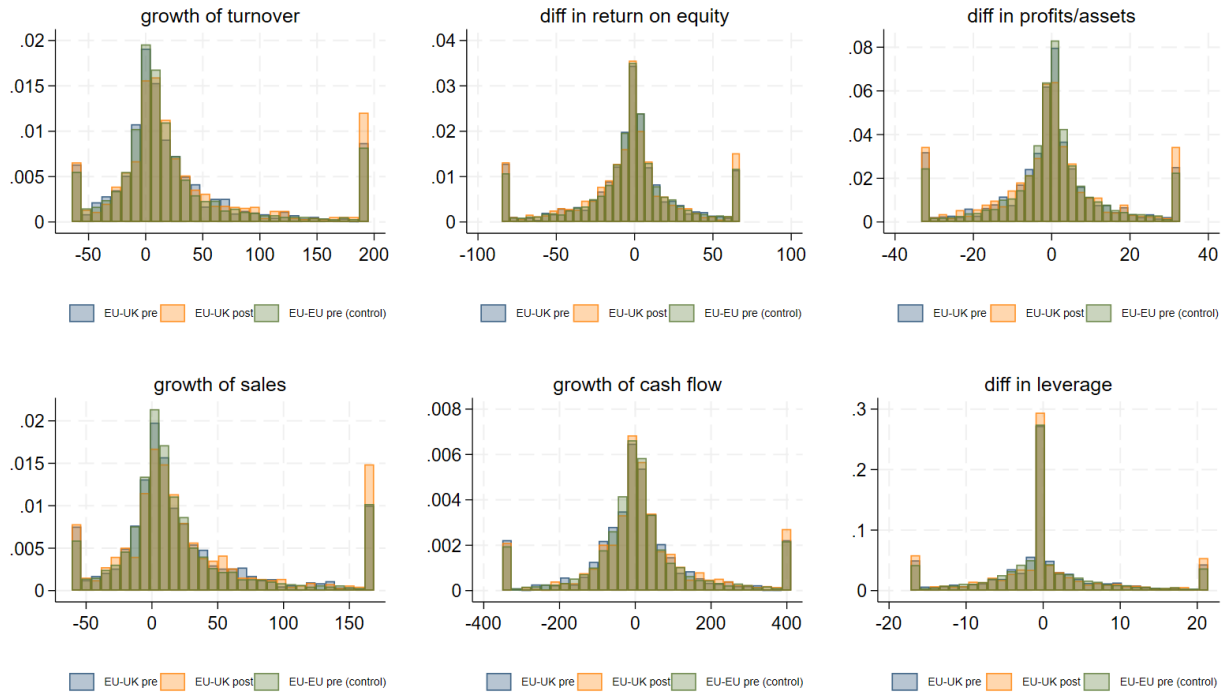
(B) Two years before the first subsidiary incorporation



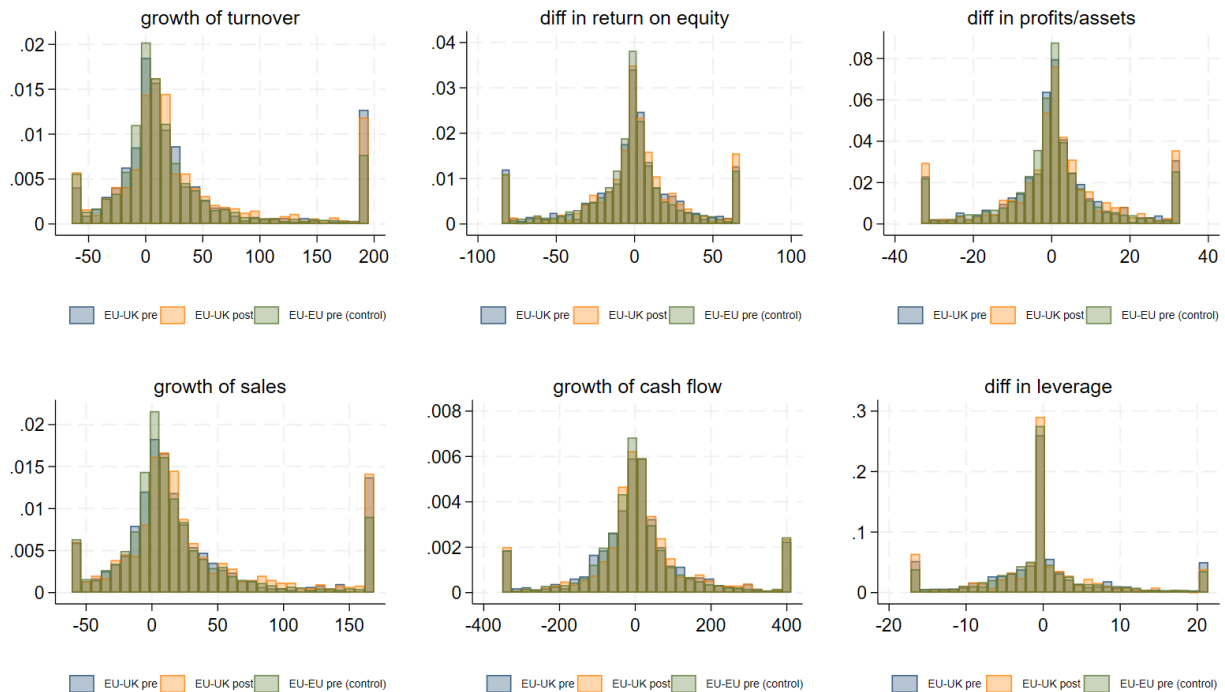
Note: Year-on-year balance sheet variables for parent firms two and one year before they establish a first subsidiary, pre and post Brexit referendum. Variables are winsorised at the 5th and 95th percentiles. Data is from *Orbis Historical*, 2011-2021.

FIGURE A.3: Balance sheet distributions of parent firms, sample EU-UK

(A) One year before the first subsidiary incorporation



(B) Two years before the first subsidiary incorporation



Note: Year-on-year balance sheet variables for parent firms two and one year before they establish a first subsidiary, pre and post Brexit referendum. Variables are winsorised at the 5th and 95th percentiles. Data is from *Orbis Historical*, 2011-2021.

TABLE A.2: Balance sheet characteristics of UK parent firms with first subsidiary, full EU sample

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-11.353*** (1.699)	-11.279*** (1.553)	-12.282*** (1.547)	-6.150*** (1.313)
Observations	15,204	16,788	12,597	13,834
R^2	0.003	0.004	0.002	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	-1.638*** (0.445)	-2.490*** (0.283)	-5.724** (2.401)	-6.771** (2.613)
Observations	16,446	18,053	14,524	15,742
R^2	0.002	0.001	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.863*** (0.167)	-0.496*** (0.123)	0.741*** (0.128)	0.274* (0.142)
Observations	18,304	20,126	15,224	16,682
R^2	0.001	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.3: Balance sheet characteristics of EU parent firms with first subsidiary, full EU sample

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	5.883* (3.774)	4.399* (2.755)	7.864** (2.869)	4.122*** (1.476)
Observations	15,559	17,257	14,275	15,726
R^2	0.003	0.004	0.003	0.003
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	3.008** (1.216)	-1.367 (1.216)	5.338 (5.485)	6.550 (6.801)
Observations	16,964	18,725	14,985	16,270
R^2	0.003	0.001	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	1.075* (0.526)	-0.598 (0.601)	-0.339* (0.184)	-0.161 (0.265)
Observations	18,792	20,764	15,418	16,941
R^2	0.001	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.4: Balance sheet characteristics of UK parent firms with first and only subsidiary

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-5.122** (1.907)	-8.529*** (1.689)	-5.301** (2.204)	-8.976*** (1.664)
Observations	7,062	7,803	5,400	5,914
R^2	0.002	0.003	0.001	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	0.144 (0.399)	-1.288*** (0.461)	2.102 (2.558)	-3.265 (3.324)
Observations	7,597	8,320	6,995	7,609
R^2	0.004	0.001	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.354 (0.216)	-0.030 (0.176)	0.875*** (0.198)	0.420* (0.228)
Observations	8,488	9,331	7,340	7,935
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.5: Balance sheet characteristics of EU parent firms with first and only subsidiary

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	11.198*** (2.964)	5.471** (2.300)	10.980*** (2.252)	3.692** (1.439)
Observations	8,242	9,160	6,789	7,478
R^2	0.002	0.002	0.004	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	4.780*** (1.319)	-1.452 (1.596)	12.989** (6.160)	7.992 (6.029)
Observations	8,069	8,930	7,400	8,085
R^2	0.006	0.002	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	1.590*** (0.548)	-0.603 (0.661)	-0.452 (0.277)	-0.158 (0.355)
Observations	8,945	9,923	7,483	8,120
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.6: Balance sheet characteristics of UK parent firms with first subsidiary, excl. 2016 and 2017

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-8.071*** (1.701)	-5.333*** (1.831)	-11.056*** (1.906)	-1.447 (1.939)
Observations	7,631	8,456	5,914	6,494
R^2	0.003	0.002	0.002	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	-1.204** (0.471)	-2.421*** (0.603)	-3.736 (3.786)	-2.861 (3.682)
Observations	8,238	9,042	7,589	8,266
R^2	0.005	0.002	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.721*** (0.193)	-0.769*** (0.258)	1.191*** (0.168)	0.291 (0.202)
Observations	9,159	10,077	7,897	8,531
R^2	0.003	0.001	0.002	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.7: Balance sheet characteristics of EU parent firms with first subsidiary, excl. 2016 and 2017

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	11.048*** (2.429)	11.001*** (2.720)	11.800*** (2.340)	7.461*** (1.391)
Observations	7,947	8,870	7,289	8,051
R^2	0.004	0.003	0.005	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	3.681*** (0.891)	-1.114 (1.295)	26.037*** (5.627)	10.108 (9.276)
Observations	8,675	9,606	7,987	8,725
R^2	0.006	0.001	0.003	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	0.940 (0.592)	-0.356 (0.691)	-0.042 (0.257)	-0.349 (0.362)
Observations	9,575	10,620	8,107	8,804
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.8: Balance sheet characteristics of UK parent firms with first subsidiary, excl. parents with unknown sector

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-4.563*** (1.593)	-8.154*** (1.667)	-7.303*** (1.828)	-4.003** (1.776)
Observations	8,685	9,592	6,603	7,220
R^2	0.002	0.003	0.002	0.001
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	-0.692** (0.332)	-2.210*** (0.437)	2.013 (2.833)	-3.958 (3.716)
Observations	9,363	10,249	8,596	9,356
R^2	0.004	0.002	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.562*** (0.197)	-0.235 (0.172)	0.917*** (0.165)	0.284 (0.215)
Observations	10,412	11,437	8,978	9,699
R^2	0.002	0.001	0.001	0.001

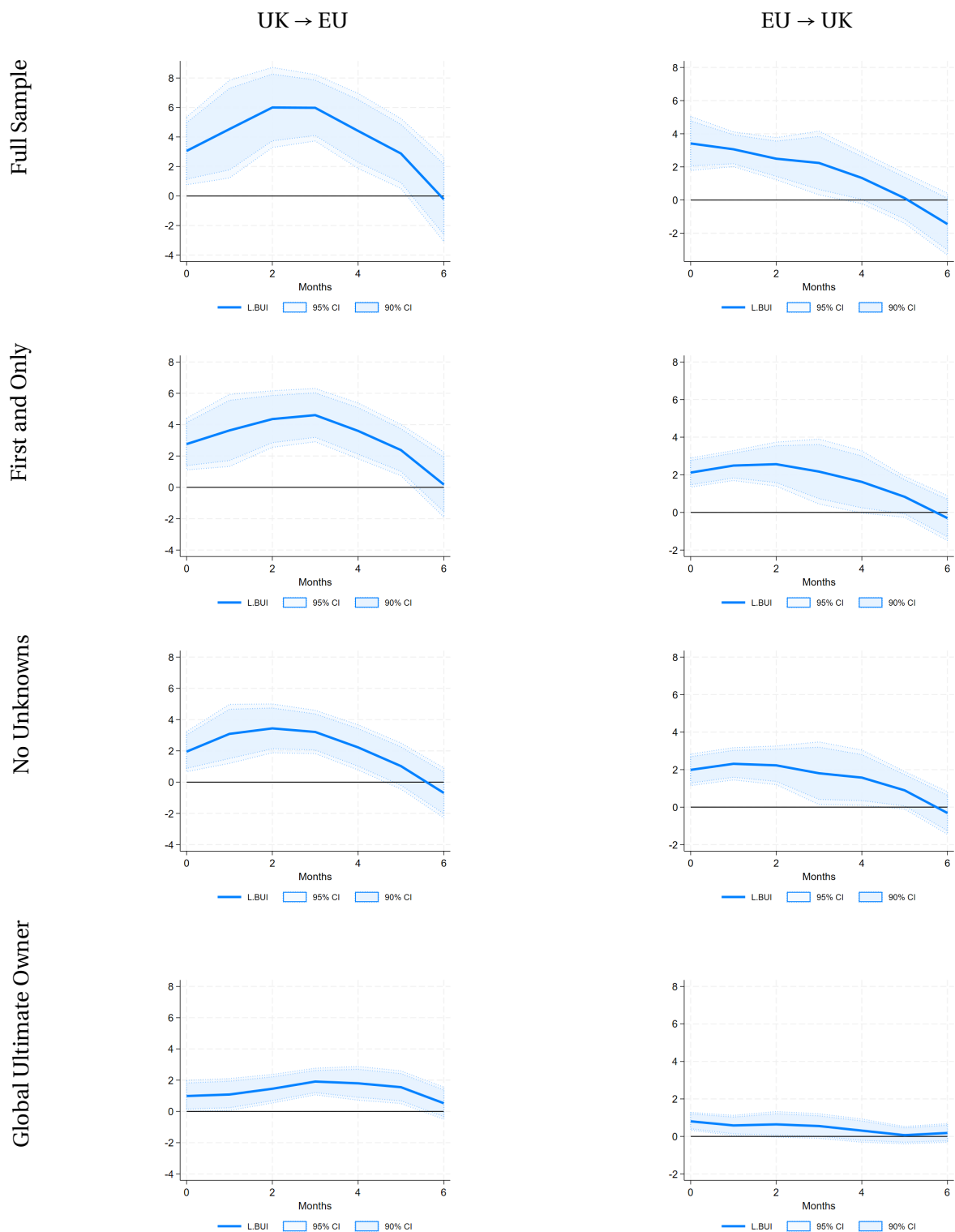
Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.9: Balance sheet characteristics of EU parent firms with first subsidiary, excl. parents with unknown sector

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	11.394*** (2.500)	8.115*** (2.287)	11.958*** (2.117)	6.508*** (1.364)
Observations	9,035	10,048	8,274	9,102
R^2	0.004	0.002	0.006	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	3.876*** (1.352)	-1.179 (1.375)	11.723** (4.941)	11.535 (7.295)
Observations	9,877	10,911	9,049	9,877
R^2	0.005	0.002	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets} \right)_{-2}$	$\Delta \left(\frac{Profits}{Assets} \right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	1.506*** (0.514)	-0.260 (0.621)	-0.209 (0.182)	-0.104 (0.328)
Observations	10,889	12,056	9,157	9,943
R^2	0.002	0.001	0.001	0.001

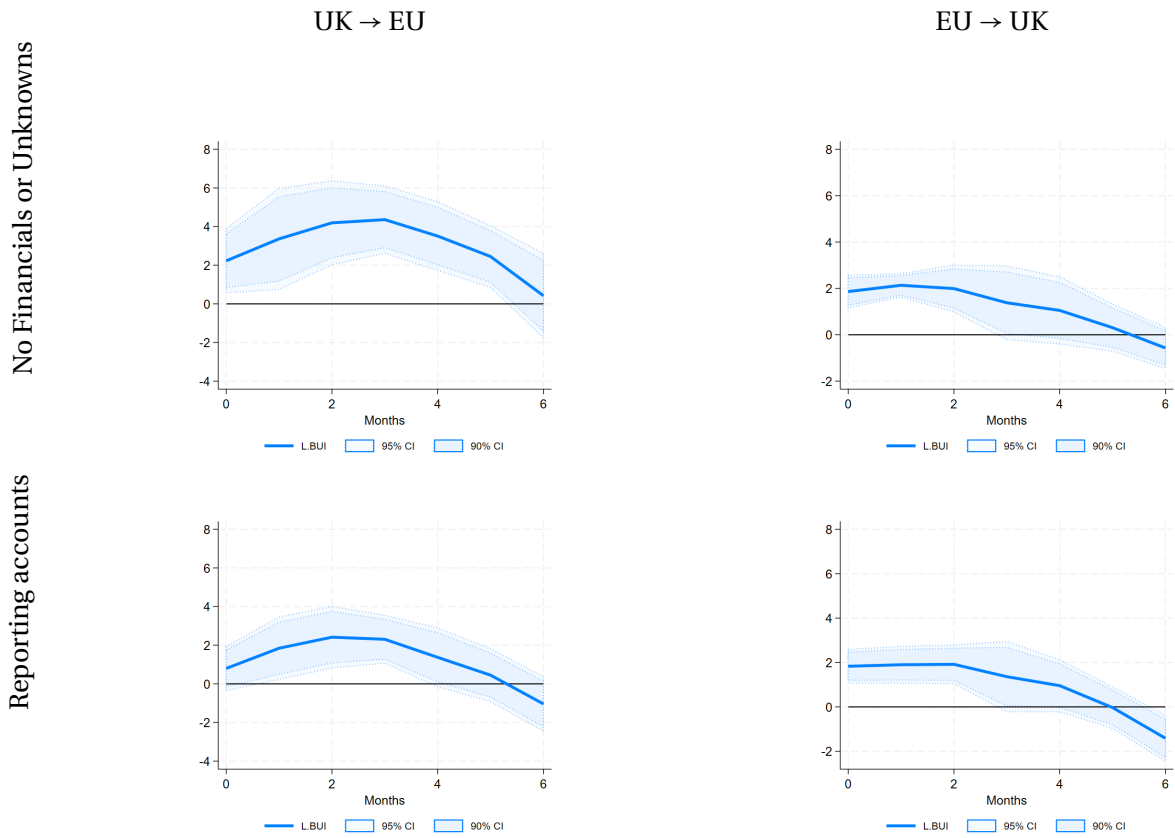
Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

FIGURE A.4: Dynamic responses of new links to uncertainty: Robustness I - Parent characteristics



Note: LPs of new links on (i) BUI_{t-1} , (ii) CUI_{t-1} , (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummy variables, (vi) Jan and Aug month dummies. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical*, *Office of National Statistics UK*, and *Eurostat*.

FIGURE A.5: Dynamic responses of new links to uncertainty: Robustness II - Subsidiary characteristics



Note: LPs of new links on (i) BUI_{t-1} , (ii) CUI_{t-1} , (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummy variables, (vi) Jan and Aug month dummies. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical*, *Office of National Statistics UK*, and *Eurostat*.

TABLE A.10: Sectors of parent firms, UK-EU and EU-UK samples

	Parent sector UK-EU sample			Parent sector EU-UK sample		
	<i>Pre-Brexit</i>	<i>Post-Brexit</i>	<i>Total</i>	<i>Pre-Brexit</i>	<i>Post-Brexit</i>	<i>Total</i>
	2011-2016	2016-2021	2011-2021	2011-2016	2016-2021	2011-2021
Accommodation	100	65	165	62	50	112
Administrative	1,459	1,151	2,610	538	552	1,090
Agriculture	19	18	37	27	34	61
Construction	159	215	374	253	204	457
Education	34	44	78	32	34	66
Electricity	65	82	147	216	67	283
Extraterritorial	33	15	48	1	0	1
Health	53	55	108	31	61	92
Household	4	6	10	0	0	0
Information	796	960	1,756	896	807	1,703
Manufacturing	559	592	1,151	818	655	1,473
Mining	90	47	137	28	20	48
Other	297	261	558	236	462	698
Professional	1,441	1,628	3,069	1,659	1,350	3,009
Public	7	0	7	2	3	5
Real estate	330	312	642	236	213	449
Recreation	65	61	126	50	58	108
Retail	616	733	1,349	889	796	1,685
Transportation	164	157	321	155	139	294
Water	17	6	23	18	20	38
Unknown	2,056	3,451	5,507	758	839	1,597
Total	8,364	9,859	18,223	6,905	6,364	13,269

Notes: Number of links in UK-EU and EU-UK by parent sector, pre and post Brexit referendum. Sectors based on NACE rev. 2 classification, and authors' own sectoral refinements, based on trade description text and company registered names. Excludes financial sector and holding companies of unknown classification. Data source *Orbis Historical*.

B. Orbis Financials Dataset

We utilize firm-level data from Bureau van Dijk's Orbis Historical, merging data on firmographics (including firm characteristics like name, age and sector) with balance sheet financials. The following sets out the precise cleaning procedure, which broadly follows [Kalemli-Ozcan et al. \(2024\)](#).

For the balance sheet variables, we aim for unique firm-financial year observations. Therefore, we first establish unique observations in terms of **BvD ID, financial year, consolidation code and filing type** by keeping the latest observation, i.e. a reporting month closer to December, and a reporting day closer to 31. Second, to get unique observations in terms of **BvD ID, year, and consolidation code**, we drop duplicate rows of these three variables where the filing type is local registry filings in favor of annual reports. This is because local registry filings are more often coded as limited financials. Third, to get unique **BvD ID and year** observations, we deal with consolidation codes. Following [Kalemli-Ozcan et al. \(2024\)](#), we keep unconsolidated over consolidated accounts to avoid double counting. Within consolidated accounts, we keep those where the company headquarter does not present an unconsolidated account. Within unconsolidated accounts, we keep those where the company headquarter does not present a consolidated account. Next, we treat same-year reporting by keeping the observation with the reporting month closer to December. Similar, for same-month reporting, we keep the reporting day closer to the end of the month. Finally, for BvD ID-year observations which differ only in terms of original reporting units, we keep the observations in "units" as this is the most frequent reporting unit.

Having eliminated duplicates, we eliminate likely reporting errors, employing basic data cleaning following [Kalemli-Ozcan et al. \(2024\)](#). As such, we drop firms with negative total assets, sales, fixed assets, tangible fixed assets, number of employees, current or non-current liabilities. Furthermore, we drop observations with more than 2 million employees. We also drop observations where total assets *and* operating revenue *and* sales *and* employment are jointly missing. Observations with a negative leverage ratio, computed as $(\text{long-term debt} + \text{loans}) / \text{total assets}$, are dropped. Finally, we construct an indicator for inconsistent balance sheets by comparing total assets to the sum of shareholder funds, current and non-current liabilities. This ratio should equal 1 for consistency, and we drop observations with a ratio below the 1st or above the 99th percentiles.